

claude-windows / df230d9b-4146-4882-a603-82c0eaffc508

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\df230d9b-4146-4882-a603-82c0eaffc508.jsonl`  
- SHA-256: `03250229a9282951c83f18ccd4079f19de4eff6d20805063c6070d6a854b9905`  
- Source modified: `2026-06-21T17:53:39+00:00`  
- Imported at: `2026-07-05T16:48:19+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `df230d9b-4146-4882-a603-82c0eaffc508`
```

Transcript

1. user (2026-06-21T17:07:19.393Z)

Implement T11 in the engine repo (badminton-highlight-indexer): a retention sweep for the engine's `output/` and `uploads/` directories, needed before non-owner (alpha) exposure of the hosted tool. Tracked in `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9 (T11) and the `ui-driven-cloud-serving-gap` thread.

Goal: a sweep (tool/job) that purges, past a retention window, the uploaded source videos (`uploads/`, written by `backend/api/uploads.py` ? default `output/uploads`) and generated artifacts (`output/` reels/frames).

IMPORTANT: do NOT hand-roll this ? there is already a cache-hygiene policy + a dry-run-first cleanup tool (`tools/cleanup\_caches.py`, plus a retention table in the cache-hygiene notes). EXTEND that tooling (or add a sibling sweep consistent with it: keep DB/CSVs/reels per the retention policy, purge stale frames/staged/uploads). Dry-run-first; loud structured logging.

OWNER-GATED: the retention WINDOW is the owner's call ? surface a recommendation (e.g. uploads 7 days, output reels per existing policy) and ask before hardcoding; default to a conservative dry-run.

Conventions (? this engine checkout is SHARED with a concurrent serving session): `git fetch origin` then `git checkout -b fix/t11-retention-sweep origin/master`; re-verify HEAD before commit; stage EXPLICIT paths (never `git add -A`); ONE PR. Full local gate before sign-off: `python run\_all\_tests.py`, `ruff check .`, `mypy` per CI, coverage ? the repo floor.

Read first: `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9, `tools/cleanup\_caches.py`, the cache-hygiene policy notes, `backend/api/uploads.py` (where uploads land), and the engine config for `security.upload\_dir`.

2. assistant (2026-06-21T17:07:26.206Z)

I'll start by reading the key files and checking repo state. Let me do the initial reconnaissance in parallel.

3. user (2026-06-21T17:07:31.887Z)

---STATUS---

On branch serving/exposure-safety-preflight

Your branch is up to date with 'origin/serving/exposure-safety-preflight'.

Untracked files:

```
(use "git add <file>..." to include in what will be committed)  
  artifacts/  
  docs/KHELSUTRA_PRODUCT_OVERVIEW.md  
  docs/PACKAGING_PLAN.md
```

nothing added to commit but untracked files present (use "git add" to track)

```

---CURRENT BRANCH---
serving/exposure-safety-preflight
---HEAD---
6fdd1d4 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed
engine (default-OFF)
---ORIGIN/MASTER---
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)

```

4. user (2026-06-21T17:07:31.930Z)

```

1      """Chunked upload (B2, PR-2) ? extracted from backend.main into its own router.
2
3      Browser uploads arrive in sequential parts ? security.max_part_mb (free-tier edge
4      proxies cap request bodies at ~100 MB ? HOSTING_PLAN $2), are assembled server-side
5      under security.upload_dir, then submitted to /api/process by the returned path.
6      Upload state is in-process by design (this is the single-box alpha): a server restart
7      aborts partial uploads and the client re-inits. Per-user partitioning of upload_dir
8      arrives with PR-3 identity scoping.
9
10     The startup config lives on ``backend.main.global_config`` (set ONLY in ``main()``).
11     ``_upload_cfg`` resolves it through the ``backend.main`` module object so it always
12     reads the live value the server set at startup (and so tests that monkeypatch
13     ``backend.main.global_config`` flow through unchanged).
14     """
15     import os
16     import threading
17     import uuid
18     from typing import Any, Dict, Optional
19
20     from fastapi import APIRouter, HTTPException, Request
21     from pydantic import BaseModel
22
23     from backend.utils.hashing import compute_video_id # canonical file-identity hashing
24     from backend.utils.paths import safe_output_filename
25
26     router = APIRouter()
27
28     _uploads: Dict[str, Dict[str, Any]] = {}
29     _uploads_lock = threading.Lock()
30
31
32     class UploadInitRequest(BaseModel):
33         filename: str
34         total_size_bytes: Optional[int] = None
35
36
37     class UploadCompleteRequest(BaseModel):
38         total_parts: int
39
40
41     def _upload_cfg():
42         # Resolved lazily (import inside the function) to read the LIVE startup config that
43         # main() sets on the module global, and to avoid a circular import at module load
44         # (main.py imports this router; this module must not import main at import time).
45         import backend.main as _main
46         sec = getattr(_main.global_config, "security", None)
47         upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
48         max_bytes = int(getattr(sec, "max_upload_mb", 4096) or 4096) * 1024 * 1024
49         part_bytes = int(getattr(sec, "max_part_mb", 95) or 95) * 1024 * 1024
50         exts = [e.lower().lstrip(".") for e in (getattr(sec, "allowed_upload_extensions",
51         None) or ["mp4", "mov"])]
52         return upload_dir, max_bytes, part_bytes, exts
53

```

```

54     def _abort_upload(upload_id: str) -> None:
55         with _uploads_lock:
56             st = _uploads.pop(upload_id, None)
57             if st:
58                 try:
59                     os.remove(st["tmp_path"])
60                 except OSError:
61                     pass
62
63
64     @router.post("/api/upload/init")
65     def upload_init(req: UploadInitRequest):
66         """Start a chunked upload. Returns upload_id; PUT sequential parts next."""
67         upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
68         try:
69             name = safe_output_filename(req.filename)
70         except ValueError as e:
71             raise HTTPException(status_code=400, detail=str(e)) from e
72         ext = os.path.splitext(name)[1].lower().lstrip(".")
73         if ext not in exts:
74             raise HTTPException(status_code=400,
75                                 detail=f"Extension '{ext}' not allowed. Allowed:
{sorted(exts)}")
76         if req.total_size_bytes is not None and req.total_size_bytes > max_bytes:
77             raise HTTPException(status_code=413,
78                                 detail=f"File exceeds the {max_bytes // (1024 * 1024)} MB
upload limit.")
79         os.makedirs(upload_dir, exist_ok=True)
80         upload_id = uuid.uuid4().hex
81         tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")
82         open(tmp_path, "wb").close()
83         with _uploads_lock:
84             _uploads[upload_id] = {"filename": name, "tmp_path": tmp_path, "parts": 0,
"bytes": 0}
85         return {"upload_id": upload_id, "max_part_mb": part_bytes // (1024 * 1024),
86                 "next_part": f"/api/upload/{upload_id}/part/0"}
87
88
89     @router.put("/api/upload/{upload_id}/part/{index}")
90     async def upload_part(upload_id: str, index: int, request: Request):
91         """Append one part (raw bytes body). Parts are strictly sequential (0,1,2,...)."""
92         _, max_bytes, part_bytes, _ = _upload_cfg()
93         with _uploads_lock:
94             st = _uploads.get(upload_id)
95             if st is None:
96                 raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
97             if index != st["parts"]:
98                 raise HTTPException(status_code=409,
99                                     detail=f"Out-of-order part: expected {st['parts']}, got
{index} (parts are sequential)")
100
101             received = 0
102             overflow = None
103             with open(st["tmp_path"], "ab") as fh:
104                 async for chunk in request.stream():
105                     received += len(chunk)
106                     if received > part_bytes:
107                         overflow = f"Part exceeds the {part_bytes // (1024 * 1024)} MB per-part
limit."
108                         break
109                     if st["bytes"] + received > max_bytes:
110                         overflow = f"Upload exceeds the {max_bytes // (1024 * 1024)} MB total
limit."
111                         break
112                     fh.write(chunk)

```

```

113         if overflow:
114             _abort_upload(upload_id)
115             raise HTTPException(status_code=413, detail=overflow + " Upload aborted ? re-
init to retry.")
116
117         with _uploads_lock:
118             st["parts"] += 1
119             st["bytes"] += received
120         return {"upload_id": upload_id, "received_parts": st["parts"], "received_bytes":
st["bytes"]}
121
122
123     @router.post("/api/upload/{upload_id}/complete")
124     def upload_complete(upload_id: str, req: UploadCompleteRequest):
125         """Finalize: verify part count, move into place, return {path, video_id} for
/api/process."""
126         upload_dir, _, _, _ = _upload_cfg()
127         with _uploads_lock:
128             st = _uploads.get(upload_id)
129             if st is None:
130                 raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
131             if st["parts"] == 0 or st["parts"] != req.total_parts:
132                 raise HTTPException(status_code=409,
133                                     detail=f"Expected {req.total_parts} part(s), received
{st['parts']}.")
134             final_path = os.path.join(upload_dir, st["filename"])
135             if os.path.exists(final_path): # never overwrite an earlier upload with the same
name
136                 stem, ext = os.path.splitext(st["filename"])
137                 final_path = os.path.join(upload_dir, f"{stem}-{upload_id[:8]}{ext}")
138             os.replace(st["tmp_path"], final_path)
139             with _uploads_lock:
140                 _uploads.pop(upload_id, None)
141             video_id = compute_video_id(final_path)
142             return {"path": os.path.abspath(final_path), "video_id": video_id,
143                     "size_bytes": os.path.getsize(final_path),
144                     "next": "POST /api/process with this path"}
145
146
147     @router.delete("/api/upload/{upload_id}")
148     def upload_abort(upload_id: str):
149         """Abort an in-flight upload and delete its partial file."""
150         with _uploads_lock:
151             known = upload_id in _uploads
152             if not known:
153                 raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
154             _abort_upload(upload_id)
155             return {"status": "aborted", "upload_id": upload_id}
156

```

5. user (2026-06-21T17:07:31.943Z)

```

1     """Cache-hygiene tool for the GPU highlight pipeline.
2
3     Scans the regenerable intermediates the pipeline leaves behind ? in the WSL stage
4     dir (`~/clips`) and the Windows `output/` tree ? classifies each entry as
5     **purge** (regenerable: frame caches, staged video copies, handoff clips, temp dirs)
6     or **keep** (durable: trajectory CSVs, the SQLite DB, final reels) and reports the
7     reclaimable space.
8
9     DRY-RUN BY DEFAULT ? it only *shows* what it would delete. Pass `--apply` to delete.
10
11     python -m tools.cleanup_caches                # report only (safe)
12     python -m tools.cleanup_caches --apply         # delete purge candidates
13     python -m tools.cleanup_caches --keep-video 89605e80ed01 # protect one video's

```

```

cache
14     python -m tools.cleanup_caches --older-than 7          # only purge entries >7 days old
15     python -m tools.cleanup_caches --include-wasbcache     # also drop detector resume
caches
16
17     Pairs with the pipeline's own self-cleaning (indexing.{wasb,tracknet}.keep_frames=false,
18     which deletes frames after a SUCCESSFUL run); this tool reclaims what failed/old runs
19     and pre-self-cleaning videos left behind. Stdlib only; safe to run anywhere (the WSL
20     scan is skipped automatically when ``wsl`` is unavailable).
21     """
22     from __future__ import annotations
23
24     import argparse
25     import os
26     import shutil
27     import subprocess
28     import sys
29     from dataclasses import dataclass
30     from typing import List, Optional, Tuple
31
32     # ---- classification (pure; unit-tested) ----- #
33
34     # (category, regenerable, default_purge). default_purge=True ? deleted unless a flag
35     # protects it. wasbcache is regenerable but default_purge=False (it enables resume and
36     # is tiny) ? opt in with --include-wasbcache.
37     _KEEP = ("keep", False, False)
38
39
40     def classify(name: str, is_dir: bool, location: str) -> Tuple[str, bool, bool]:
41         """Classify a cache entry by name. Returns (category, regenerable,
default_purge)."""
42         low = name.lower()
43         if is_dir:
44             if low.endswith("_frames"):
45                 return ("frame cache", True, True)
46             if low.endswith("_wasbcache"):
47                 return ("detector resume cache", True, False) # opt-in via --include-
wasbcache
48             if low.startswith("chunks_") and low.endswith("_handoff"):
49                 return ("ai handoff clips", True, True)
50             if low.startswith("tmp"):
51                 return ("temp dir", True, True)
52             return ("other dir", False, False)
53         # files
54         if low.endswith((".csv",)):
55             return ("trajectory csv", False, False)          # durable analysis output
56         if low.endswith((".db", ".sqlite", ".sqlite3")):
57             return ("database", False, False)                # cached rallies ? never purge
58         if low.endswith((".log", ".sh", ".txt", ".json", ".md")):
59             return ("log/script", False, False)
60         if low.endswith((".mp4", ".avi", ".mov", ".mkv")):
61             # In the WSL stage dir, every video is a regenerable staged copy.
62             if location == "wsl":
63                 return ("staged video", True, True)
64             # In output/, a 1080p/proxy is regenerable; anything else is a produced reel.
65             if "_1080p" in low or "_proxy" in low or low.endswith("_proxy.mp4"):
66                 return ("proxy video", True, True)
67             return ("highlight reel", False, False)          # durable output ? keep
68         return ("other", False, False)
69
70
71     @dataclass
72     class Entry:
73         name: str
74         path: str          # display path (WSL ~/clips/<name> or windows abs)

```

```

75         size: int            # bytes
76         mtime: float         # epoch seconds (0 if unknown)
77         is_dir: bool
78         location: str         # "wsl" | "windows"
79
80     @property
81     def category(self) -> Tuple[str, bool, bool]:
82         return classify(self.name, self.is_dir, self.location)
83
84
85     def plan_deletions(entries: List[Entry], *, keep_video: Optional[str], older_than_days:
float,
86                        include_wasbcache: bool, now: float) -> Tuple[List[Entry],
List[Entry]]:
87         """Split entries into (to_delete, kept) per the flags. Pure ? no I/O."""
88         to_delete: List[Entry] = []
89         kept: List[Entry] = []
90         cutoff = older_than_days * 86400.0
91         for e in entries:
92             cat, _regen, default_purge = e.category
93             purge = default_purge or (include_wasbcache and cat == "detector resume cache")
94             if purge and keep_video and keep_video in e.name:
95                 purge = False
96             if purge and older_than_days > 0 and e.mtime > 0 and (now - e.mtime) < cutoff:
97                 purge = False
98             (to_delete if purge else kept).append(e)
99         return to_delete, kept
100
101
102     # ---- scanning (I/O) ----- #
103
104     def human(n: int) -> str:
105         f = float(n)
106         for unit in ("B", "K", "M", "G", "T"):
107             if f < 1024 or unit == "T":
108                 return f"{f:.1f}{unit}" if unit != "B" else f"{int(f)}B"
109             f /= 1024
110         return f"{f:.1f}T"
111
112
113     def _wsl_lines(distro: str, cmd: str) -> List[str]:
114         """Run a bash command in WSL and return stdout lines ([]) on any failure."""
115         try:
116             res = subprocess.run(["wsl", "-d", distro, "bash", "-lc", cmd],
117                                 capture_output=True, text=True, timeout=300)
118         except (OSError, subprocess.TimeoutExpired):
119             return []
120         return (res.stdout or "").splitlines()
121
122
123     def scan_wsl(distro: str, stage_dir: str) -> List[Entry]:
124         """List ~/clips entries with recursive size + mtime + type. [] if WSL absent.
125
126         Uses ``du -sb`` (recursive byte sizes, native-fast on ext4) merged with ``find
127         -printf`` (type + mtime) by basename. Deliberately avoids a ``for`` loop ? the
128         user's login shell mangles loop variables, but du/find arg-globbing is reliable.
129         """
130         if not shutil.which("wsl"):
131             return []
132         size_lines = _wsl_lines(distro, f"du -sb {stage_dir}/* {stage_dir}/.[!.*]
2>/dev/null || true")
133         meta_lines = _wsl_lines(distro, f"find {stage_dir}/ -maxdepth 1 -mindepth 1 "
134                                f"-printf '%y\\tT@\\t%f\\n' 2>/dev/null || true")
135         meta = {} # basename -> (is_dir, mtime)
136         for line in meta_lines:

```

```

137         parts = line.split("\t")
138         if len(parts) == 3:
139             y, mt, name = parts
140             try:
141                 meta[name] = (y == "d", float(mt))
142             except ValueError:
143                 continue
144     entries: List[Entry] = []
145     for line in size_lines:
146         if "\t" not in line:
147             continue
148         sz, path = line.split("\t", 1)
149         name = path.rsplit("/", 1)[-1]
150         try:
151             size = int(sz)
152         except ValueError:
153             continue
154         is_dir, mtime = meta.get(name, (False, 0.0))
155         entries.append(Entry(name=name, path=f"{stage_dir}/{name}", size=size,
156                             mtime=mtime, is_dir=is_dir, location="wsl"))
157     return entries
158
159
160 def _dir_size(path: str) -> int:
161     total = 0
162     for root, _dirs, files in os.walk(path):
163         for fn in files:
164             try:
165                 total += os.path.getsize(os.path.join(root, fn))
166             except OSError:
167                 pass
168     return total
169
170
171 def scan_windows(output_dir: str) -> List[Entry]:
172     """List output/ entries (top level) plus the output/wasb sub-tree."""
173     entries: List[Entry] = []
174     for base in (output_dir, os.path.join(output_dir, "wasb")):
175         if not os.path.isdir(base):
176             continue
177         for de in os.scandir(base):
178             # 'wasb' is scanned as its own base below ? don't also list it as a dir
179             here.
180             if base == output_dir and de.name == "wasb":
181                 continue
182             try:
183                 is_dir = de.is_dir()
184                 size = _dir_size(de.path) if is_dir else de.stat().st_size
185                 mtime = de.stat().st_mtime
186             except OSError:
187                 continue
188             entries.append(Entry(name=de.name, path=de.path, size=size,
189                                 mtime=mtime, is_dir=is_dir, location="windows"))
190     return entries
191
192 def _delete(e: Entry, distro: str) -> bool:
193     try:
194         if e.location == "wsl":
195             subprocess.run(["wsl", "-d", distro, "bash", "-lc", f'rm -rf {_q(e.path)}'],
196                             capture_output=True, text=True, timeout=600, check=True)
197         elif e.is_dir:
198             shutil.rmtree(e.path, ignore_errors=True)
199         else:
200             os.remove(e.path)

```

```

201         return True
202     except (OSError, subprocess.SubprocessError):
203         return False
204
205
206 def _q(path: str) -> str:
207     if path.startswith("~/"):
208         return "~/'" + path[2:].replace("'", "'\\'") + "'"
209     return "'" + path.replace("'", "'\\'") + "'"
210
211
212 def _report(title: str, rows: List[Entry]) -> int:
213     print(f"\n{title}")
214     if not rows:
215         print(" (none)")
216         return 0
217     total = 0
218     for e in sorted(rows, key=lambda x: x.size, reverse=True):
219         cat = e.category[0]
220         print(f" {human(e.size):>8} {cat:<22} {e.path}")
221         total += e.size
222     print(f" {'-' * 8}")
223     print(f" {human(total):>8} TOTAL ({len(rows)} items)")
224     return total
225
226
227 def main(argv: Optional[List[str]] = None) -> int:
228     ap = argparse.ArgumentParser(description="Report/clean regenerable pipeline caches
229 (dry-run by default).")
230     ap.add_argument("--apply", action="store_true", help="actually delete (default:
231 report only)")
232     ap.add_argument("--keep-video", default=None, help="protect entries whose name
233 contains this id/substring")
234     ap.add_argument("--older-than", type=float, default=0.0, help="only purge entries
235 older than N days")
236     ap.add_argument("--include-wasbcache", action="store_true", help="also purge
237 detector resume caches")
238     ap.add_argument("--stage-dir", default="~/clips", help="WSL stage dir (default
239 ~/clips)")
240     ap.add_argument("--distro", default="Ubuntu", help="WSL distro (default Ubuntu)")
241     ap.add_argument("--output-dir", default="output", help="Windows output dir (default
242 output)")
243     ap.add_argument("--summary", action="store_true",
244 help="print a single-line nudge (only if reclaimable >= --threshold-
245 gb); for hooks")
246     ap.add_argument("--threshold-gb", type=float, default=5.0,
247 help="--summary stays silent below this many reclaimable GB (default
248 5)")
249     args = ap.parse_args(argv)
250
251     import time
252     now = time.time()
253     entries = scan_wsl(args.distro, args.stage_dir) + scan_windows(args.output_dir)
254     to_delete, kept = plan_deletions(
255         entries, keep_video=args.keep_video, older_than_days=args.older_than,
256         include_wasbcache=args.include_wasbcache, now=now)
257
258     if args.summary:
259         reclaim = sum(e.size for e in to_delete)
260         if reclaim >= args.threshold_gb * (1024 ** 3):
261             print(f"?? {human(reclaim)} of regenerable pipeline cache reclaimable "
262 f"? run `python -m tools.cleanup_caches` to review, `--apply` to
263 delete.")
264     return 0
265

```



```

256     reclaim = _report("PURGE candidates (regenerable):", to_delete)
257     _report("KEEP (durable / protected):", kept)
258     print(f"\nReclaimable: {human(reclaim)}")
259
260     if not args.apply:
261         print("\nDRY RUN ? nothing deleted. Re-run with --apply to delete the PURGE
list.")
262         return 0
263     if not to_delete:
264         print("\nNothing to delete.")
265         return 0
266
267     print(f"\nDeleting {len(to_delete)} items...")
268     freed, failed = 0, 0
269     for e in to_delete:
270         if _delete(e, args.distro):
271             freed += e.size
272         else:
273             failed += 1
274             print(f"  FAILED: {e.path}")
275     print(f"Freed ~{human(freed)}" + (f" ({failed} failed)" if failed else ""))
276     return 0
277
278
279     if __name__ == "__main__":
280         sys.exit(main())
281

```

6. assistant (2026-06-21T17:07:36.013Z)

Now let me read the architecture doc §9, the config, the cache-hygiene memory, and existing tests for the cleanup tool.

7. user (2026-06-21T17:07:38.094Z)

```

===== T11 section of LOCAL_APPLIANCE_ARCHITECTURE.md =====
grep: docs/LOCAL_APPLIANCE_ARCHITECTURE.md: No such file or directory

```

8. user (2026-06-21T17:07:38.859Z)

```

<system-reminder>This memory is 8 days old. Memories are point-in-time observations, not live
state ? claims about code behavior or file:line citations may be outdated. Verify against
current code before asserting as fact.</system-reminder>
1     ---
2     name: cache-hygiene-policy
3     description: "Retention policy + tooling for the GPU pipeline's regenerable caches
(frames/staged videos) ? what's purgeable vs durable, the self-cleaning runners,
tools/cleanup_caches.py, disk guard, and the SessionStart nudge hook."
4     metadata:
5         node_type: memory
6         type: project
7         originSessionId: 47ecf23a-6312-4118-b2bd-c1af33f2eaa7
8     ---
9
10    How we keep pipeline caches from accumulating (designed + shipped 2026-06-13, PR #109).
11    Pairs with [[gotcha-long-runs-gpu-wsl]] (why frames balloon: ~93 GB/4K video).
12
13    **Principle:** separate **regenerable intermediates** (purge aggressively,
automatically)
14    from **durable outputs** (keep/archive). Purging is default-safe ? dry-run first.
15
16    **Retention table:**
17    - PURGE (regenerable): `*_frames/` (PNG decode), staged `*.mp4` copies in `~/clips`,
18      `*_1080p`/proxy videos in `output/`, `chunks*_handoff/`, `tmp*/`.

```

```

19 - KEEP (durable): trajectory CSVs (WASB outputs / golden corpus), the SQLite rally DB,
20 final highlight reels, logs/scripts.
21 - OPT-IN purge: `*_wasbcache/` (detector resume cache ? small, aids resume; only via
22 `--include-wasbcache`).
23
24 **Mechanisms (all live on master):**
25 1. **Self-cleaning runners (the root fix):** `indexing.{wasb,tracknet}.keep_frames`
26 (default **false**) ? after a SUCCESSFUL run (CSV written) the WASB runner deletes
the
27 frame cache + staged video; TrackNet deletes its staged video. On FAILURE nothing is
28 deleted (resume intact). So per-video frame bloat no longer accumulates by default.
29 2. **`tools/cleanup_caches.py`** ? dry-run-first ops tool. `python -m
tools.cleanup_caches`
30 (report), `--apply` (delete), `--keep-video <id|stem>` (protect; pass the STEM e.g.
31 `GX010125` to also catch the Windows proxy, the hash only matches WSL frames),
32 `--older-than N`, `--include-wasbcache`, `--summary` (one-line nudge). Scans WSL
`~/clips`
33 + Windows `output/`. ? WSL scan uses `du`+`find` arg-globbing on purpose ? the
owner's
34 login shell mangles `for e in *` loops (loop var comes back EMPTY); `du -sh ./` /
35 `find -printf` are reliable. wsl.exe output decodes fine with text=True here (not
UTF-16).
36 3. **Disk guard:** `indexing.{wasb,tracknet}.min_free_gb` (default 0=off) ? warns before
a
37 run if WSL free space is low.
38 4. **SessionStart hook** (`claude/settings.local.json`, local+gitignored): runs
39 `python -m tools.cleanup_caches --summary` ? one-line nudge only when >5 GB
reclaimable.
40
41 **Manual cadence:** review the dry-run at session wrap / after each completed video;
delete
42 frame caches once the trajectory CSV + DB rallies exist. Keep `keep_frames=false` (the
default)
43 unless debugging a specific run.
44

```

9. user (2026-06-21T17:07:39.362Z)

```

tests\test_cleanup_caches.py
tests\__pycache__\test_cleanup_caches.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\bhi-execpolicy\tests\test_cleanup_caches.py
.claude\worktrees\bhi-
execpolicy\tests\__pycache__\test_cleanup_caches.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\test_cleanup_caches.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\__pycache__\test_cleanup_caches.cpython-313-
pytest-9.0.3.pyc
.claude\worktrees\tier1-windowing\tests\test_cleanup_caches.py
.claude\worktrees\tier1-windowing\tests\__pycache__\test_cleanup_caches.cpython-313-pytest-
9.0.3.pyc
.claude\worktrees\w2-safeguard\tests\test_cleanup_caches.py
.claude\worktrees\w2-safeguard\tests\__pycache__\test_cleanup_caches.cpython-313-pytest-
9.0.3.pyc
.claude\worktrees\hardchop\tests\test_cleanup_caches.py
.claude\worktrees\hardchop\tests\__pycache__\test_cleanup_caches.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\boxhill17\tests\test_cleanup_caches.py
.claude\worktrees\pr-stitch\tests\test_cleanup_caches.py
.claude\worktrees\pr-stitch\tests\__pycache__\test_cleanup_caches.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\gap-doc\tests\test_cleanup_caches.py
.claude\worktrees\resweep\tests\test_cleanup_caches.py
.claude\worktrees\recdoc\tests\test_cleanup_caches.py

```

10. assistant (2026-06-21T17:07:42.926Z)

The architecture doc has a different name. Let me find it and read the existing test file.

11. user (2026-06-21T17:07:45.102Z)

```
=== docs with APPLIANCE/PACKAGING/T11/retention ===
docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs/COMPUTE_DECOUPLED_SERVING/README.md
docs/PACKAGING_PLAN.md
=== glob appliance ===
PACKAGING_PLAN.md
VIDEO_LOCALITY_MODEL.md
```

12. user (2026-06-21T17:07:45.412Z)

```
1      """Unit tests for the cache-hygiene tool's pure logic (classify + plan_deletions).
2
3      The I/O (wsl/du/find scan, deletion) is not unit-tested here ? these guard the
4      classification + filtering rules that decide what gets deleted, which is the part
5      where a mistake would be destructive.
6      """
7      from tools.cleanup_caches import Entry, classify, plan_deletions
8
9
10     def test_classify_regenerable_intermediates():
11         assert classify("GX010125_1080p__abc_frames", True, "wsl")[2] is True      # frame
cache -> purge
12         assert classify("GX010125_1080p__abc.mp4", False, "wsl")[2] is True      #
staged video -> purge
13         assert classify("GX010125_1080p.mp4", False, "windows")[2] is True      # proxy
-> purge
14         assert classify("chunks_gemini_handoff", True, "windows")[2] is True      #
handoff clips -> purge
15         assert classify("tmp9xk2", True, "windows")[2] is True                  # temp
dir -> purge
16
17
18     def test_classify_durable_kept():
19         assert classify("adarsh_traj.csv", False, "wsl")[2] is False              #
trajectory -> keep
20         assert classify("sports_indexer.db", False, "windows")[2] is False      # DB ->
keep
21         assert classify("GX010125_highlights.mp4", False, "windows")[2] is False  # reel
-> keep
22         assert classify("run_wasb.sh", False, "wsl")[2] is False                  #
script -> keep
23         # wasbcache is regenerable but opt-in only (default_purge False)
24         cat, regen, default_purge = classify("adarsh_traj_wasbcache", True, "wsl")
25         assert cat == "detector resume cache" and regen is True and default_purge is False
26
27
28     def _e(name, size, is_dir, location="wsl", mtime=0.0):
29         return Entry(name=name, path=f"~/clips/{name}", size=size, mtime=mtime,
30                       is_dir=is_dir, location=location)
31
32
33     def test_plan_default_purges_frames_keeps_csv_and_wasbcache():
34         entries = [
35             _e("v_frames", 90_000, True),
36             _e("v.mp4", 2_000, False),
37             _e("v_traj.csv", 100, False),
38             _e("v_traj_wasbcache", 500, True),
39         ]
40         to_del, kept = plan_deletions(entries, keep_video=None, older_than_days=0,
41                                       include_wasbcache=False, now=1_000_000.0)
42         names = {e.name for e in to_del}
43         assert names == {"v_frames", "v.mp4"}
44         assert {e.name for e in kept} == {"v_traj.csv", "v_traj_wasbcache"}
```

```

45
46
47 def test_plan_include_wasbcache_opts_it_in():
48     entries = [_e("v_traj_wasbcache", 500, True)]
49     to_del, _ = plan_deletions(entries, keep_video=None, older_than_days=0,
50                               include_wasbcache=True, now=1_000_000.0)
51     assert [e.name for e in to_del] == ["v_traj_wasbcache"]
52
53
54 def test_plan_keep_video_protects_matching_substring():
55     entries = [_e("GX010125_frames", 90_000, True), _e("kushagra_frames", 5_000, True)]
56     to_del, kept = plan_deletions(entries, keep_video="GX010125", older_than_days=0,
57                                   include_wasbcache=False, now=1_000_000.0)
58     assert [e.name for e in to_del] == ["kushagra_frames"]
59     assert [e.name for e in kept] == ["GX010125_frames"]
60
61
62 def test_plan_older_than_protects_recent():
63     now = 1_000_000.0
64     entries = [
65         _e("recent_frames", 90_000, True, mtime=now - 3600),      # 1h old
66         _e("old_frames", 90_000, True, mtime=now - 10 * 86400),    # 10d old
67     ]
68     to_del, kept = plan_deletions(entries, keep_video=None, older_than_days=7,
69                                   include_wasbcache=False, now=now)
70     assert [e.name for e in to_del] == ["old_frames"]
71     assert [e.name for e in kept] == ["recent_frames"]
72

```

13. assistant (2026-06-21T17:07:49.613Z)

Let me find where T11 §9 is actually documented.

14. user (2026-06-21T17:07:51.689Z)

```

===== COMPUTE_DECOUPLED_SERVING/README.md T11 context =====
20: - (a) Hosted service ? operate on small local uploads (<2GB) and load from
gdrive / a bucket, spinning up Cloud Run for both rally segmentation AND stitching
(stitching is compute-intensive ? run it where its cost is accounted for).
37:| D10 | Free-tier "data-for-reels" trade: free reels in exchange for training rights
(uploaded footage + derived trajectories/labels); paid tier = no-train (privacy is the paid
feature). Carve-outs: train ONLY on attested ADULT footage (minor footage ? reel yes,
training pool NO ? DPDP/GDPR need verifiable parental consent); train on DERIVED data +
auto-delete raw; ToS counsel-reviewed; live training-on-uploads gated to M9 (public
signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties [[paper-coauthor-aim]] /
PERSONALIZATION flywheel. |
55:| ADR-011 | DECOUPLED scope = M0?M2-M3.5; deferred M3 (serving) + M4 (billing/cost-
guard); overrode the "rent nothing / not a service" posture | This project realizes
M3/M4. Carries §06's owner-gated gates: DPA/consent for non-owner uploads, collector `md5`
keying, WASB-C3. |
99:2. Reliably stores them durably in the bucket (the consented-retention bucket; raw video
kept only for the training pool, not the no-consent path).
102:This is the consent?capture?eval?goldenize flywheel (the cloud realization of the
PERSONALIZATION contribution model): every consented free reel makes the owned model better.
Reuses `data_pipeline/`, `backend/eval/`, and the M9 consent gate; adults-only, raw-
retention gated by the D10 consent.
146:## 9. Definition of done
155:ADR lineage: ADR-005/008/009/010/011/012/013 ? ADR-014
(`docs/archives/decisions/DECISIONS.md`). Strategy: `PLATFORM_ARCHITECTURE.md`,
`HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`,
`LOCAL_APPLIANCE_ARCHITECTURE.md` (khelsutra). Archived predecessor:
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-
ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). Code
anchors `[verified]`: backend/pipeline/detectors/base.py:164, tracknet_runner.py:281,
stitcher/compiler.py:303, segmenters/trajectory_hybrid.py:65,237, backend/storage/`,

```

```
`backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`,  
`backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-  
mappers ? synthesis; 58 findings).
```

15. user (2026-06-21T17:07:52.399Z)

```
===== PACKAGING_PLAN.md T11 context =====  
34:| **L10** | **North star = the FULL quality path is the SOLE completion goal** ? LIGHT is the  
first increment; DONE requires native WASB + BYO training + in-app annotation | 02, 2026-06-21 |  
$9 DoD splits "v1 shippable" from "north-star DONE"; P3/P4 (WASB/ONNX/train/annotate) are  
completion-gating, not optional. |  
112:2. **02 ? RESOLVED 2026-06-21:** v1 = **LIGHT first**, **but the sole completion goal (north  
star, L10) is the FULL path** ? native WASB + BYO training + in-app annotation. DoD $9 splits  
"v1 shippable" from "north-star DONE"; P3/P4 are completion-gating, not optional.  
121:## 9. Definition of done  
154:- 2026-06-21 ? Owner resolved 01 + 02: **cross-platform (Win/Mac/Linux) webserver, no native  
component** (L9); **LIGHT-first with a WASB-complete north star as the sole completion goal**  
(L10). Updated $0/$2/$4.5/$6/$7-P2b/$8/$9 accordingly. Remaining gates: 03 (WASB-C3 counsel), 04  
(hosted vs BYO-cloud), 05?09.
```

16. assistant (2026-06-21T17:09:25.872Z)

The `LOCAL\_APPLIANCE\_ARCHITECTURE.md` lives in the sibling khelsutra repo (referenced as "(khelsutra)"), not this engine ? the task requirements are self-contained though. Let me read the security config, the decoupled-serving \$9 context, and the CI gate config.

17. user (2026-06-21T17:09:27.097Z)

```
===== security config (upload_dir) =====  
backend/config/models.py:280:class SecurityConfig(BaseModel):  
backend/config/models.py:288:     # contain upload_dir or /api/process will reject the uploaded  
paths.  
backend/config/models.py:289:     upload_dir: str = "output/uploads"  
backend/config/models.py:292:     max_upload_mb: int = 4096  
backend/config/models.py:294:     allowed_upload_extensions: List[str] = ["mp4", "mov"]  
Binary file backend/config/__pycache__/models.cpython-313.pyc matches
```

18. user (2026-06-21T17:09:29.451Z)

```
===== CI: ruff/mypy/coverage floor =====  
.github/workflows/ci.yml:16: # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle +  
bugbear.  
.github/workflows/ci.yml:17: # Also the repo-wide syntax guard ? ruff reports invalid-syntax on  
every  
.github/workflows/ci.yml:28:     - name: Install ruff  
.github/workflows/ci.yml:29:       run: python -m pip install ruff  
.github/workflows/ci.yml:31:       run: ruff check . --output-format github  
.github/workflows/ci.yml:33: # Type gate (mypy.ini at repo root): lenient green baseline with  
an explicit  
.github/workflows/ci.yml:35: # up. Typed core deps are installed so mypy sees their real  
signatures  
.github/workflows/ci.yml:45:     - name: Install mypy + typed core deps  
.github/workflows/ci.yml:46:       run: python -m pip install mypy pydantic fastapi uvicorn  
python-dotenv google-genai numpy  
.github/workflows/ci.yml:48:       run: mypy backend training  
.github/workflows/ci.yml:87:     # (pytest, pytest-cov, httpx for the fastapi TestClient,  
ruff, mypy).  
.github/workflows/ci.yml:97:       run: python run_all_tests.py --cov=backend --cov=training  
--cov-report=term --cov-fail-under=70  
=== pyproject/setup.cfg mypy+ruff ===  
pyproject.toml:4:# (no `backend/__init__.py`; mypy.ini sets explicit_package_bases for the same  
pyproject.toml:45:     "ruff",  
pyproject.toml:46:     "mypy",  
pyproject.toml:84:[tool.hatch.metadata]
```

19. user (2026-06-21T17:09:30.188Z)

```

90     ### 4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
91     DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In
    **cloud-serving, stitch is co-located in the Cloud Run job** so `gpu_active_seconds` (detect) +
    `wall_seconds` (detect+stitch) + `bytes_out` (reel egress) land on **one metered invocation** ?
    the per-job cost ledger. Running stitch on a laptop would leave that cost
    **invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in
    the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get
    bursty ? the job table + `claim_due_job` already supports both.)
92
93     ### 4.5 Reuse map ? most of the seams already exist `[verified]`
94     **Exists:** the 3-stage core; `_resolve_runner` (the single compute seam);
    `StubDetectorRunner` (`replay_csv` = byte-exact $0 anchor); `StorageBackend` ABC + 4 backends +
    `StorageRef.parse_uri`; the worker CLI (fetch/put, `--config`/`--set`); the async job control
    plane (`jobs` table, `claim_due_job` CAS); `ExecutionPolicy WAIT_AND_RESUME`/`JobWaiting`
    (carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB`); `skip_ai_handoff`
    (LocalNoAI); storage fakes; golden fixtures + experiment harness. **Partial:** per-video
    workspace helper (default-OFF); `DeviceContext` (designed); edge-auth (v4 `user_id` columns
    exist). **Net-new:** the `deployment` section; `input_backend`; configurable output base (kill
    hardcoded `output/`); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-
    auth header extraction; startup env detection.
95
96     ### 4.6 The data-flywheel harness (D12 / M11) `[design]`
97     Gemini-on (D11) ships good reels now; this harness is **how we earn the right to flip
    Gemini off**. On a **consented, adult** cloud-serving job (the D10 trade), the pipeline
    additionally:
98     1. **Captures** the source upload + the Gemini reel + the rally output
    (intervals/report) into the per-video workspace (`workspaces/<video_id>/`), instead of the no-
    consent auto-delete path.
99     2. **Reliably stores** them durably in the bucket (the consented-retention bucket; raw
    video kept only for the training pool, not the no-consent path).
100    3. **Runs shadow evals** ? the owned model scores the same video alongside Gemini
    (`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks
    the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").
101    4. **Promotes to golden** ? captured videos that pass review become **golden TRAINING
    rows** (human-reviewed labels via the annotation flow ? `data_pipeline/`), growing the corpus ?
    the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.
102    This is the consent?capture?eval?goldenize **flywheel** (the cloud realization of the
    PERSONALIZATION contribution model): every consented free reel makes the owned model better.
    Reuses `data_pipeline/`, `backend/eval/`, and the M9 consent gate; **adults-only**, raw-
    retention gated by the D10 consent.
103
104    ## 6. Honest limits ? what this does NOT do / prove
105    - A green CI suite proves **plumbing + determinism (window+stitch) + profile-parity**
    for $0 ? it does **NOT** prove field quality, nor that real-GPU detection is good (that's owner-
    side, real-video).
106    - **Cross-GPU detection is NOT byte-identical** ? parity is a window-level claim, not a
    CSV-byte claim.
107    - The cloud `0.1.0-l4` weights are **n=2 plumbing-grade** ? cloud-serving ships with
    **Gemini handoff ON** until the owned model clears a floor (open Q).
108    - This does not resolve **WASB-C3** (sharing a trained model) or the **DPA/consent**
    gate ? both remain owner/counsel-gated before public, non-owner serving.
109
110    ## 7. Deliverables & progress tracker ? **source of truth**
111    Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off
    `origin/master`, no stacking. Default-OFF where it touches core callers.
112
113    | ID | Deliverable | Depends | Gated? | Status | PR |
114    |----|-----|-----|-----|-----|----|
115    | M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? |
    **? APPROVED 2026-06-21** |
116    | M1 | `deployment.profile` + `compute_target` config + preset bundles +

```

```

precedence/auto-detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? |
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
117 | M2 | `input_backend`/`input_root`; wire main CLI/API input through
`StorageRef.parse_uri` (local=identity); storage-fake tests | M1 | safe | ? |
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
118 | M3 | Configurable output/scratch base ? kill hardcoded `output/`
(`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden
+ stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |
119 | M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 |
safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283) |
120 | M5 | **truly-local profile** end-to-end (preset + startup checks); first committed
**runlog** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |
121 | M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +
**containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | **cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch
on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) |
M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds,
wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation +
`/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission
? owner/follow-up) |
124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`;
DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? |
[#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT
verify + owner_id tag, default-OFF ?; **exposure-safety boot posture** in `startup.py` ? edge-
auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see
changelog; consent cols + ToS/DPA ? owner/counsel) |
125 | M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit
owner approval | ? | ? |
126 | **M11** | **Data-flywheel harness (D12):** on a consented adult job, **capture** the
upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-
eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the
golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the
owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent
gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | ? |
127 | **M12** | **gdrive-api** (OAuth) StorageBackend (D14 Northstar):** cloud reads the
source from + writes the stitched reel **back to the user's Google Drive next to the source**
(per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local
mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | ? |
128
129 **Testing strategy is a first-class deliverable** ?
[`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ?
[runlogs/](runlogs/).
130
131 ## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
132 **? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:**
133 1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
134 3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
135 4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to
proxy).
136 5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data;
counsel-gated to M9).
137 6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until
the owned model clears the floor; capture?store?eval?goldenize meanwhile).
138
139 **? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:**
140 2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the
job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).

```

```

141      7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the
source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ?
distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
142      8. ? **Codec ? D16:** **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only**
(speed, no output-byte change); NVENC-encode deferred behind a config knob.
143      9. ? **Input cap ? D16:** **hard-reject`<2GB`** for v1, as a config cap (raise to 4GB
later in one line).
144      10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the
name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.
145
146      ## 9. Definition of done
147      - [ ] M1?M4 land (config + input + output-base + device layer), each green on golden
cross-OS + experiment no-regression.
148      - [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical
windows + segment ranges.
149      - [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog /
DEPLOY_REPORT** with a real sample run.

```

20. user (2026-06-21T17:09:44.435Z)

```

275      # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
276      # them to WARNING; set true to restore full chatter for request-flow debugging.
277      verbose_http: bool = False
278
279
280      class SecurityConfig(BaseModel):
281          # When set, /api/process video_path must resolve under this directory (hosted/tenant
282          # mode). Default None preserves the single-user local 'any local file' workflow.
283          allowed_video_root: Optional[str] = None
284
285          # --- Chunked upload (B2, PR-2) ---
286          # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
287          # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
288          # contain upload_dir or /api/process will reject the uploaded paths.
289          upload_dir: str = "output/uploads"
290          # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
291          # free-tier edge proxies cap request bodies there (HOSTING_PLAN §2).
292          max_upload_mb: int = 4096
293          max_part_mb: int = 95
294          allowed_upload_extensions: List[str] = ["mp4", "mov"]
295
296          # --- M9 edge auth (Cloudflare Access, D9) ---
297          # DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local
single-user
298          # path is unchanged). When True, /api/process requires + verifies the `Cf-Access-
Jwt-Assertion`
299          # header against the team JWKS (so a direct hit to the Cloud Run URL can't spoof the
identity
300          # header) and tags the job with the verified user_id (owner_id). The team domain
(e.g.
301          # ``https://<team>.cloudflareaccess.com``) supplies the JWKS + issuer; the AUD is
the CF Access
302          # application's audience tag. NO secrets here ? these are public identifiers.
303          edge_auth_enabled: bool = False
304          cf_team_domain: str = ""          # e.g. https://<team>.cloudflareaccess.com (issuer +
/certs JWKS)
305          cf_access_aud: str = ""          # the CF Access application AUD tag the token must
carry
306          # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS env var ? read at
import in
307          # backend/main.py so importing the app does no filesystem I/O; default [""].
Credentials stay
308          # OFF ? Cf-Access is a header, not a cookie.)
309
310

```



```

311     class StorageConfig(BaseModel):
312         """Pluggable storage backend (docs/DECOUPLED_COMPUTE/). Default ``local`` = today's
313         behaviour (filesystem paths, byte-for-byte). ``s3`` covers AWS + Cloudflare R2 + DO
Spaces
314         + MinIO (via ``endpoint_url``); ``gcs`` = Google Cloud Storage; ``gdrive`` = a
locally MOUNTED

```

21. user (2026-06-21T17:09:45.236Z)

```

80         - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
81         # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
82         # need an out-of-band index that PEP 621 can't express. Keep installing
83         # them separately here, before the editable install pulls the rest.
84         run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
85         - name: Install package (editable) + dev tooling
86         # Editable PEP 621 install: runtime deps + the `dev` group
87         # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
88         run: python -m pip install -e .[dev]
89         # obsreport (private, soft dep) is absent here: reporting tests skip via
90         # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
91         # is therefore calibrated against THIS lane's number (local runs with
92         # obsreport installed measure a little higher).
93         # Floor calibrated 2026-06-11 against this lane's measured 72% (local
94         # with obsreport: 73%). Ratchet it UP as coverage grows; never down
95         # without an owner decision.
96         - name: Run full suite (with coverage floor)
97         run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
98
99         # Cross-OS determinism guard for the committed golden regression fixtures

```

22. assistant (2026-06-21T17:10:07.803Z)

I now have full context. Key facts:

- `tools/cleanup\_caches.py` already sweeps `output/` (frames/proxies/staged) with dry-run-first + `--older-than`, but treats `output/uploads` as an opaque "other dir" ? ****never swept**** (no retention on uploaded source videos).
- Uploads land flat in `security.upload\_dir` (default `output/uploads`), as final `` videos plus orphaned `.partial-<id>` temp files from aborted uploads.
- CI gate: `ruff check .`, `mypy backend training` (tool isn't type-checked), coverage floor 70 on `backend`+`training` (tool isn't measured ? but I'll still add unit tests per repo convention).

The retention ****window**** is the owner-gated decision. Let me confirm it before baking a default.

23. user (2026-06-21T17:10:39.067Z)

Your questions have been answered: "Default retention window for uploaded source videos in output/uploads (purge older than N days)? The sweep stays dry-run-first regardless ? this only sets the default cutoff (overridable via --uploads-retention-days).="7 days (Recommended)", "Should the sweep also purge generated output reels past a window, or keep them durably per the existing cache-hygiene policy?="Keep reels durably (Recommended)". You can now continue with these answers in mind.

24. assistant (2026-06-21T17:10:44.786Z)

Owner confirmed: ****uploads window default = 7 days; keep reels durably per policy.**** Now let me set up the branch from `origin/master`.

25. user (2026-06-21T17:10:46.092Z)

branch 'fix/t11-retention-sweep' set up to track 'origin/master'.
Switched to a new branch 'fix/t11-retention-sweep'

---VERIFY HEAD---
fix/t11-retention-sweep
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)

26. assistant (2026-06-21T17:10:55.588Z)

Let me verify the files on this branch match what I read (the serving branch may have diverged on these paths).

27. user (2026-06-21T17:10:57.641Z)

=== files changed between origin/master and the serving branch in our target paths ===
(empty above = identical, my earlier reads are valid)

28. assistant (2026-06-21T17:12:04.313Z)

Owner confirmed. Now implementing. Let me mark the chapter and extend the cleanup tool. First the docstring + classification:

29. user (2026-06-21T17:12:05.124Z)

Chapter marked: "T11 retention sweep". Continue your current work.

30. user (2026-06-21T17:12:23.993Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

31. assistant (2026-06-21T17:12:28.434Z)

Now the `classify` function ? add the uploads-location branch:

32. user (2026-06-21T17:12:34.177Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

33. assistant (2026-06-21T17:12:37.253Z)

Now `plan\_deletions` ? add a per-location retention window for uploads, and update the `Entry.location` comment:

34. user (2026-06-21T17:12:38.620Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

35. user (2026-06-21T17:12:52.022Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

36. assistant (2026-06-21T17:12:55.723Z)

Now `scan\_windows` to skip the nested uploads dir, plus a new `scan\_uploads`:

37. user (2026-06-21T17:13:06.875Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

38. assistant (2026-06-21T17:13:11.460Z)

Now the `main` function ? add args, wire the uploads scan, retention banner, and `--json` structured output:

39. user (2026-06-21T17:13:42.557Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

40. assistant (2026-06-21T17:13:50.346Z)

I used `Dict` in the new code ? let me add it to the imports:

41. user (2026-06-21T17:13:51.423Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

42. assistant (2026-06-21T17:14:23.413Z)

Now let me add unit tests for the new uploads logic. Let me append to the existing test file:

43. user (2026-06-21T17:14:41.480Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

44. assistant (2026-06-21T17:14:46.476Z)

Now let me verify: run the focused tests, ruff on the changed files, and a CLI smoke test.

45. user (2026-06-21T17:14:49.243Z)

```
..... [100%]  
12 passed in 0.12s
```

46. user (2026-06-21T17:14:49.575Z)

```
          | ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^  
370 |          + (f"purge > {args.older_than:g}d old" if args.older_than > 0 else "purge all  
regenerable (no age guard)")  
371 |      print(f"  KEEP (durable, never purged): highlight reels, the rally DB, trajectory  
CSVs, logs/scripts")  
          |  
help: Remove extraneous `f` prefix  
  
F541 [*] f-string without any placeholders  
--> tools\cleanup_caches.py:371:11  
          |  
369 |      print(f"  caches (frames/staged/proxies): "  
370 |          + (f"purge > {args.older_than:g}d old" if args.older_than > 0 else "purge all  
regenerable (no age guard)")  
371 |      print(f"  KEEP (durable, never purged): highlight reels, the rally DB, trajectory  
CSVs, logs/scripts")  
          |  
          | ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^  
372 |  
373 |      reclaim = _report("PURGE candidates:", to_delete)  
          |  
help: Remove extraneous `f` prefix  
  
Found 3 errors.  
[*] 3 fixable with the `--fix` option.
```

47. assistant (2026-06-21T17:14:55.493Z)

Three F541 lint errors (f-strings without placeholders). Let me fix them:

48. user (2026-06-21T17:14:59.692Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

49. assistant (2026-06-21T17:15:04.593Z)

Now re-run ruff and do an end-to-end CLI smoke test against a throwaway temp tree:

50. user (2026-06-21T17:15:05.967Z)

All checks passed!
=== ruff exit: 0 ===

51. user (2026-06-21T17:15:09.680Z)

Exit code 1
===== DEFAULT DRY-RUN REPORT =====
C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe: Error while finding module specification for 'tools.cleanup_caches' (ModuleNotFoundError: No module named 'tools')

52. assistant (2026-06-21T17:15:14.267Z)

The `cd` broke module resolution. Let me run from the repo root pointing `--output-dir` at the temp tree:

53. user (2026-06-21T17:15:24.588Z)

```
===== DEFAULT DRY-RUN (uploads window 7d) =====
Retention windows in effect:
  uploads (C:/Users/avidu/AppData/Local/Temp/tmp.SZ5dJHG8p7/output/uploads): purge source videos
> 7d old
  caches (frames/staged/proxies): purge all regenerable (no age guard)
  KEEP (durable, never purged): highlight reels, the rally DB, trajectory CSVs, logs/scripts

PURGE candidates:
  10.0M frame cache
C:/Users/avidu/AppData/Local/Temp/tmp.SZ5dJHG8p7/output/GX_frames
  2.0M uploaded source
C:/Users/avidu/AppData/Local/Temp/tmp.SZ5dJHG8p7/output/uploads/old_match.mp4
  4.0K partial upload
C:/Users/avidu/AppData/Local/Temp/tmp.SZ5dJHG8p7/output/uploads/.partial-deadbeef
-----
  12.0M TOTAL (3 items)
  .   10.0M frame cache (1)
  .    2.0M uploaded source (1)
  .    4.0K partial upload (1)

KEEP (durable / protected):
  1.0M uploaded source
C:/Users/avidu/AppData/Local/Temp/tmp.SZ5dJHG8p7/output/uploads/recent_match.mp4
  488.3K highlight reel
C:/Users/avidu/AppData/Local/Temp/tmp.SZ5dJHG8p7/output/GX_highlights.mp4
-----
  1.5M TOTAL (2 items)

Reclaimable: 12.0M

DRY RUN ? nothing deleted. Re-run with --apply to delete the PURGE list.
TMPDIR=/tmp/tmp.SZ5dJHG8p7
```

54. assistant (2026-06-21T17:15:30.916Z)

The retention sweep works exactly as designed: 9-day-old upload + orphan partial ? purge, fresh upload + reel ? kept, frames ? purge, no double-listing. Now let me verify `--json` and `--apply`:

55. user (2026-06-21T17:15:36.547Z)

```
===== --json PLAN =====
{
  "dry_run": true,
  "windows": {
    "uploads_retention_days": 7.0,
    "cache_older_than_days": 0.0
  },
  "uploads_dir": "C:/Users/avidu/AppData/Local/Temp/tmp.fklgIThH0L/output\\uploads",
  "reclaimable_bytes": 12587008,
  "purge": [
    {
      "path": "C:/Users/avidu/AppData/Local/Temp/tmp.fklgIThH0L/output\\GX_frames",
      "category": "frame cache",
      "location": "windows",
      "size_bytes": 10485760,
      "age_days": 0.0
    },
    {
      "path": "C:/Users/avidu/AppData/Local/Temp/tmp.fklgIThH0L/output\\uploads\\old_match.mp4",
      "category": "uploaded source",
      "location": "uploads",
      "size_bytes": 2097152,
      "age_days": 9.0
    },
    {
      "path": "C:/Users/avidu/AppData/Local/Temp/tmp.fklgIThH0L/output\\uploads\\.partial-
deadbeef",
      "category": "partial upload",
      "location": "uploads",
      "size_bytes": 4096,
      "age_days": 9.0
    }
  ],
  "by_category": [
    {
      "category": "frame cache",
      "count": 1,
      "bytes": 10485760
    },
    {
      "category": "uploaded source",
      "count": 1,
      "bytes": 2097152
    },
    {
      "category": "partial upload",
      "count": 1,
      "bytes": 4096
    }
  ]
}
===== --apply (loud per-item) =====
```

Reclaimable: 12.0M

Deleting 3 items...

PURGED 10.0M frame cache

62. user (2026-06-21T17:16:14.981Z)

```
=== is artifacts/ tracked? ===
(empty = untracked, not in any PR)
=== ruff on everything EXCEPT untracked artifacts ===
ruff (excl artifacts) exit=0
```

63. assistant (2026-06-21T17:16:19.967Z)

Confirmed clean. Now the full test suite with the CI coverage gate:

64. user (2026-06-21T17:19:07.674Z)

backend\sports\badminton.py	13	1	92%
backend\sports\base.py	12	3	75%
backend\storage__init__.py	57	4	93%
backend\storage\base.py	14	0	100%
backend\storage\gcs.py	41	4	90%
backend\storage\gdrive.py	61	3	95%
backend\storage\local.py	49	3	94%
backend\storage\ref.py	48	0	100%
backend\storage\s3.py	47	5	89%
backend\tools__init__.py	0	0	100%
backend\tools\export.py	75	2	97%
backend\tools\rally_slicer.py	104	47	55%
backend\utils__init__.py	0	0	100%
backend\utils\app_home.py	40	0	100%
backend\utils\geometry.py	29	0	100%
backend\utils\hashing.py	9	0	100%
backend\utils\logging_setup.py	11	0	100%
backend\utils\paths.py	33	3	91%
backend\utils\redact.py	16	0	100%
backend\utils\run_telemetry.py	126	12	90%
backend\utils\single_instance.py	43	11	74%
backend\utils\video.py	25	3	88%
backend\utils\workspace.py	49	0	100%
backend\worker__init__.py	0	0	100%
backend\worker__main__.py	294	37	87%
training__init__.py	0	0	100%
training\gen0__init__.py	0	0	100%
training\gen0\harness.py	129	9	93%

TOTAL	9748	1433	85%

Required test coverage of 70% reached. Total coverage: 85.30%
1290 passed, 7 warnings in 163.23s (0:02:43)

=====

RESULT: PASS - all tests green

=====

65. assistant (2026-06-21T17:19:27.308Z)

Full gate is green:

- **ruff**: clean on all tracked files (the 2 errors are in untracked `artifacts/`, not in this PR).
- **mypy**: `backend training`: `Success: no issues found in 134 source files`.
- **suite**: `1290 passed`, 0 failed.
- **coverage**: 85.30% ? above the 70% floor, no regression (changes are in `tools/`+`tests/`, outside the measured `backend/`+`training` scope).

Now commit ? re-verifying HEAD and staging explicit paths only:

66. user (2026-06-21T17:19:31.022Z)

```
=== re-verify branch + HEAD (must be mine) ===
```

```
fix/t11-retention-sweep
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)
=== confirm origin/master hasn't moved under me ===
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)
=== staging EXPLICIT paths only ===
M tests/test_cleanup_caches.py
M tools/cleanup_caches.py
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
?? docs/PACKAGING_PLAN.md
```

67. assistant (2026-06-21T17:19:34.744Z)

Only my two files staged; `origin/master` unchanged (no sibling commit in the gap). Committing:

68. user (2026-06-21T17:19:41.968Z)

```
[fix/t11-retention-sweep 7940355] feat(serving): T11 ? retention sweep for stale uploads +
caches (extends cleanup_caches, dry-run-first)
 2 files changed, 225 insertions(+), 26 deletions(-)
=== verify only my commit is ahead of origin/master ===
7940355 feat(serving): T11 ? retention sweep for stale uploads + caches (extends cleanup_caches,
dry-run-first)
```

69. assistant (2026-06-21T17:20:00.999Z)

Committed cleanly ? `origin/master..HEAD` lists only my commit. Let me update the cache-hygiene memory to reflect the extended tool (its retention table + flags changed):

70. user (2026-06-21T17:20:08.125Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\cache-hygiene-policy.md has been updated successfully. (file state is current in your context ? no need to Read it back)

71. user (2026-06-21T17:20:21.157Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\cache-hygiene-policy.md has been updated successfully. (file state is current in your context ? no need to Read it back)

72. assistant (2026-06-21T17:20:44.096Z)

Done. T11 retention sweep is implemented, verified, and committed on `fix/t11-retention-sweep` (not pushed ? holding per the "don't push/open a PR unless asked" convention).

What I built

Rather than hand-rolling, I **extended** `tools/cleanup\_caches.py` (the existing dry-run-first cache-hygiene tool) to also sweep the upload landing zone ? keeping one consistent retention surface.

```
 **[tools/cleanup_caches.py](tools/cleanup_caches.py)**
- New `uploads` scan location for `security.upload_dir` (default `output/uploads`).
  `classify()` tags uploaded source videos and orphaned partial-<id> files (aborted/in-flight chunks) as purge candidates; a future per-user subdir (PR-3) is left untouched.
- `scan_windows()` now skips the nested uploads dir so it isn't double-counted as an opaque "other dir" ? which had silently shielded uploads from any retention.
- `plan_deletions()` gained an independent `uploads_retention_days` window (privacy-driven), orthogonal to the cache --older-than` guard (disk-driven). `0` = off.
- Loud structured output: a retention-window banner, per-category subtotals, per-item
```


`PURGED`/`FAILED` lines on `--apply`, and a `--json` plan for a cron/job wrapper. ****Dry-run by default****; `--apply` required to delete. Reels / DB / CSVs stay durable per policy.

****tests/test_cleanup_caches.py**** ? +6 tests: uploads classification, retention protect-recent/purge-old, window independence from the cache guard, `window=0` off, category subtotals, and the nested-dir skip (a real-filesystem safety test).

****Owner gate honored**** uploads ****7d****, reels ****kept durable**** (your 2026-06-21 answer) baked as the `--uploads-retention-days` default; both overridable.

Verified locally (full gate)

- `ruff check .` ? clean on all tracked files ? *(the only 2 ruff errors are in untracked `artifacts/landing/make\_og.py`, another session's WIP ? not in this PR; CI won't see it)*
- `mypy backend training` ? `Success: no issues found in 134 source files` ?
- `python run\_all\_tests.py` ? ****1290 passed****, 0 failed ?
- Coverage ****85.30%**** (floor 70; no regression ? changes are in `tools/`+`tests/`, outside the measured scope) ?
- Manual smoke test on a temp tree confirmed: 9-day-old upload + orphan partial ? purged; fresh upload + reel ? kept; `--json` and `--apply` both correct.

Two notes:

- ****Tracking doc**** T11 is tracked in `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md §9`, which lives in the ****khelsutra sibling repo**** (not this engine checkout), so I couldn't update it from here. I did update the `cache-hygiene-policy` memory to reflect the new sweep.
- ****Not automated yet**** this is the manual/dry-run-first tool + a `--json` plan hook. Wiring it into a scheduled job (cron / Cloud Run sidecar) for live alpha is a natural follow-up ? I left the `--json` surface ready for it but didn't add scheduling (out of scope for one PR).

Want me to ****push `fix/t11-retention-sweep` and open the PR****?

73. user (2026-06-21T17:21:05.603Z)

yes please open the PRs and ensure the coverage has not regressed

74. assistant (2026-06-21T17:22:30.935Z)

Pushing and opening the PR now. These are serialized, outward-facing git operations on a checkout shared with the concurrent serving session ? so I'm doing them directly and carefully rather than fanning out subagents (parallel git writes on a shared checkout would violate the serialize-repo-writes rule). I'll kick off the master-baseline coverage measurement in parallel since that part is safely isolatable.

Let me first confirm the remote and push:

75. user (2026-06-21T17:22:32.473Z)

```
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (fetch)
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (push)
=== gh auth ===
github.com
  ? Logged in to github.com account avidullu (keyring)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_*****
```

76. assistant (2026-06-21T17:22:37.759Z)

Remote confirmed (`Khelsutra/badminton-highlight-indexer`), gh authed. Pushing the branch:

77. user (2026-06-21T17:22:40.473Z)

```
remote:
remote: Create a pull request for 'fix/t11-retention-sweep' on GitHub by visiting:
remote: https://github.com/Khelsutra/badminton-highlight-
```

```
indexer/pull/new/fix/t11-retention-sweep
remote:
branch 'fix/t11-retention-sweep' set up to track 'origin/fix/t11-retention-sweep'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      fix/t11-retention-sweep -> fix/t11-retention-sweep
```

78. assistant (2026-06-21T17:22:55.885Z)

Branch pushed. Now I'll kick off the **master-baseline coverage** in an isolated worktree (background, safely read-only w.r.t. the main checkout) while I open the PR in the foreground:

79. user (2026-06-21T17:22:58.171Z)

```
Preparing worktree (detached HEAD c039b90)
HEAD is now at c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-
flight job state (default-OFF) (#298)
=== worktree ready at origin/master ===
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)
```

80. assistant (2026-06-21T17:23:00.814Z)

Now launch the baseline coverage run in that worktree (background) and write the result to a file I can read when it finishes:

81. user (2026-06-21T17:23:02.573Z)

Command running in background with ID: buzsmkrhz. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\df230d9b-4146-4882-a603-82c0eaffc508\tasks\buzsmkrhz.output. You will be notified when
it completes. To check interim output, use Read on that file path.

82. assistant (2026-06-21T17:23:11.849Z)

Baseline running. Now opening the PR:

83. user (2026-06-21T17:23:28.190Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/299>

84. assistant (2026-06-21T17:23:34.222Z)

PR opened: **#[299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)**. While the baseline coverage runs, let me nail down the structural non-regression proof and check CI status:

85. user (2026-06-21T17:23:36.176Z)

```
=== diff vs origin/master touches ONLY tools/ + tests/ ===
tests/test_cleanup_caches.py | 78 +++++
tools/cleanup_caches.py      | 173 ++++++-----
2 files changed, 225 insertions(+), 26 deletions(-)

=== does the changed tool import backend/training? (empty = no) ===
(none ? tool is stdlib-only, pulls in zero measured-scope code)
```

86. assistant (2026-06-21T17:23:41.991Z)

Structural proof is airtight: **zero `backend/`/`training/` lines changed**, and the tool imports nothing from the measured scope. Let me check the PR's CI status and peek at the baseline run:

87. user (2026-06-21T17:23:45.063Z)

=== PR #299 CI checks ===

Golden fixtures cross-OS (macos-latest) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359167>
Mypy (lenient baseline) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359255>
Syntax sweep + import smoke (no GPU stack) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359150>
Ruff lint pass 10s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359197>
Built wheel smoke (install + launch) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359147>
Dependency license scan (no copyleft in runtime deps) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359349>
Full test suite (offline, mocked) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359181>
Golden fixtures cross-OS (ubuntu-latest) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359179>
Golden fixtures cross-OS (windows-latest) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359161>

=== baseline run progress (tail) ===

=====
.....

88. assistant (2026-06-21T17:24:00.662Z)

CI is running ? **Ruff lint already passed in CI** (confirming the untracked `artifacts/` errors are a non-issue, as predicted). Let me watch CI to completion while the baseline finishes:

89. user (2026-06-21T17:26:27.073Z)

Golden fixtures cross-OS (ubuntu-latest) pass 25s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359179>
Golden fixtures cross-OS (windows-latest) pass 1m2s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359161>
Built wheel smoke (install + launch) pass 1m21s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359147>
Dependency license scan (no copyleft in runtime deps) pass 32s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359349>
Full test suite (offline, mocked) pass 2m41s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359181>
Golden fixtures cross-OS (macos-latest) pass 17s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359167>
Golden fixtures cross-OS (ubuntu-latest) pass 25s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359179>
Golden fixtures cross-OS (windows-latest) pass 1m2s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359161>
Mypy (lenient baseline) pass 31s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359255>
Ruff lint pass 10s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359197>
Syntax sweep + import smoke (no GPU stack) pass 19s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359150>
Built wheel smoke (install + launch) pass 1m21s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27911904077/job/82590359147>

```
Dependency license scan (no copyleft in runtime deps)    pass    32s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27911904077/job/82590359349
Full test suite (offline, mocked)                      pass    2m41s  https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27911904077/job/82590359181
Golden fixtures cross-OS (macos-latest)                pass    17s    https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27911904077/job/82590359167
Golden fixtures cross-OS (ubuntu-latest)               pass    25s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27911904077/job/82590359179
Golden fixtures cross-OS (windows-latest)              pass    1m2s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27911904077/job/82590359161
Mypy (lenient baseline) pass    31s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27911904077/job/82590359255
Ruff lint pass    10s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27911904077/job/82590359197
Syntax sweep + import smoke (no GPU stack)             pass    19s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27911904077/job/82590359150
=== watch exit=0 ===
```

90. assistant (2026-06-21T17:26:31.165Z)

All ****CI checks on PR #299 are green**** ?, and the baseline coverage run finished. Let me read the baseline TOTAL and compare:

91. user (2026-06-21T17:26:32.619Z)

```
=== origin/master BASELINE coverage ===
TOTAL                                9748    1456    85%
Required test coverage of 70% reached. Total coverage: 85.06%
1277 passed, 7 skipped, 7 warnings in 144.95s (0:02:24)
```

92. assistant (2026-06-21T17:27:04.201Z)

****Coverage did not regress ? confirmed.**** Let me clean up the baseline worktree:

93. user (2026-06-21T17:27:06.179Z)

```
removed worktree (exit=0)
C:/Users/avidu/Projects/badminton-highlight-indexer
7940355 [fix/t11-retention-sweep]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
8736e2d [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gap-doc
f04f3f3 [docs/rally-gap-closure]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/pr-stitch
a6adcc5 [fix/runner-stitch-watermark]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/recdoc
9fd4215 [docs/recording-single-match]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/resweep
fe34013 [docs/lovo-resweep-n13]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
```

94. assistant (2026-06-21T17:27:31.347Z)

Worktree cleaned up. Everything is done and verified. Let me record this checkpoint in the live current-state memory (per the working-agreement: refresh on PR push):

95. user (2026-06-21T17:27:32.240Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4      only on request). What just landed, what's next, open decisions."
5      metadata:
6          node_type: memory
7          type: project
8          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
9      ---
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
13     PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
14     state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
15     don't let them pile up here.
16
17     **Checkpoint:** 2026-06-21 (**UX A-items + North-Star UPLOAD ? 4 khelsutra PRs
18     merged**). The UX/localization track (separate session from the serving/alpha-shareable one
19     below). Shipped the SPA UX A-items + the runnable-tool keystone. **4 PRs merged ? all green CI,
20     browser-verified, each adversarially reviewed via a workflow:**
21     [#61](https://github.com/avidullu/khelsutra/pull/61) **A4** (?44px mobile tap targets + tooltip-
22     text?visible: the "not available yet" reason + per-card shot confidence) .
23     [#62](https://github.com/avidullu/khelsutra/pull/62) **A5** (ONE shared `lib/errorText.ts` +
24     `errors.` namespace across build+load+ingest ? no more raw `ApiError`/`HTTP 500`; logs gain
25     code+detail) . [#63](https://github.com/avidullu/khelsutra/pull/63) **A8** (background-
26     processing reassurance + icon+text `StatusChip`, a "Stitching your reel?" building card, ??
27     status icons, aria-live) . **[#66](https://github.com/avidullu/khelsutra/pull/66) T4 ? in-UI
28     file UPLOAD (NORTH-STAR keystone)** (file picker ? chunked `/api/upload/` via new
29     `api/client.ts` helpers + `hooks/useUpload.ts` ? server path ? `/api/process` ? existing
30     poll?load?compile?download; **153 web tests**); rebased onto the sibling's #64/#65 ?
31     `apiBase()`+upload coexist cleanly). **Owner decisions (AskUserQuestion):** build T4 first; **B3
32     catalog-home = hoist to repo-root `/locales/` (LOCKED)**; record/spin-off the rest. **Spun off
33     as owner-launchable chips:** B3 (`task_b3060842`), MULTILINGUAL Phase A es/da/id
34     (`task_e15e9643`), T11 retention (`task_2480b9e2`) ? ? B3+MULTILINGUAL both edit catalogs, don't
35     run together. **Remaining:** A9 (tap-only filters), A3 (recording-edu kn/hi), native review of
36     AI-drafted kn/hi(+es/da/id); **T2 = owner stand-up only** (sibling shipped the deploy kit #65) +
37     the hosted-mode `allowed_video_root` ? `upload_dir` caveat (from the T4 review). ? Caught+fixed
38     a slip: started T4 edits on `master`, moved to a feature branch BEFORE any commit. ? Track
39     handoff = [[next-session-ux-localization]]. [[working-agreement-merging]] [[gotcha-shared-
40     checkout-branch-collisions]]
41
42     15
43     16     **Checkpoint:** 2026-06-21 (**ALPHA-SHAREABLE ? Claude's half DONE; 3 PRs merged**).
44     Owner's #1 = "highlights from a page I can share, no SSH." **KEY CORRECTION (anti-spiral
45     [[lesson-verify-engine-surface-before-scoping]]):** the icebreaker said do "B-P2 ingest screen +
46     M7 Cloud Run *service*" ? but a recon **workflow** (`wf_5de56b64`, 5 readers across both repos)
47     proved **B-P2 was ALREADY shipped** (khelsutra #56, live e2e `DRY_RUN_CUJ.md` §9) + CORS is
48     already env-driven ? **alpha-shareable is a DEPLOY/AUTH task, not an SPA build**. **Owner
49     decisions (2x AskUserQuestion):** topology **i) Tunnel-to-a-box**, sub-topology **SPLIT** (SPA
50     on CF Pages + engine on `api.` cloudflared tunnel + CF Access; single-origin Caddy kept as the
51     alt). **3 PRs MERGED:** ? **engine [#294](https://github.com/Khelsutra/badminton-highlight-
52     indexer/pull/294)** (`0c0e064`) **exposure-safety boot posture** ? `startup.py` fails fast at
53     boot (server-only `include_exposure=True`) when `edge_auth_enabled` is ON but half-configured
54     (`EXPOSED_NO_VIDEO_ROOT`/`EXPOSED_EDGE_AUTH_INCOMPLETE` fatal, `EXPOSED_AUTH_EXTRA_MISSING`
55     warn); **default-OFF**); aligned `auth.resolve_tenant` whitespace; adversarial 3-lens review
56     (`wf_35b735a9`) ? no bypass; +16 tests; 1259 green cross-OS. ? **khelsutra
57     [#64](https://github.com/avidullu/khelsutra/pull/64)** (`8bbc557`) **SPA `VITE_API_BASE`** ?
58     `apiBase()` env-driven (default same-origin `/api`) + `vite-env.d.ts` + 3 tests; build-smoke
59     proved the override bakes in. ? **khelsutra
60     [#65](https://github.com/avidullu/khelsutra/pull/65)** (`4627830`) **deploy kit** ?
```

```

`deploy/DEPLOY_ALPHA.md` (split runbook + CORS-preflight gotcha + posture test) +
`cloudflared.config.example.yml` + tracker pointers. Each isolated/default-OFF/adversarial-
reviewed/CI-green/merged-on-green; khelsutra built in a dedicated **worktree** (main tree had
sibling A8 WIP, since merged #63; ? `rmdir` the `node_modules` junction BEFORE `git worktree
remove`). **? REMAINING for alpha = OWNER stand-up only** (cloudflared tunnel + CF Pages + CF
Access + key rotation per `khelsutra/deploy/DEPLOY_ALPHA.md`). **? NEXT engine code (Claude-
doable, batch-authorized): M12** (gdrive-api, clean, no DB migration) then **M11** (flywheel
skeleton, privacy-gated on v7 consent). [[ui-driven-cloud-serving-gap]] [[lesson-verify-engine-
surface-before-scoping]]
17
18 **?? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session) ? ?
SUPERSEDED by the checkpoint above (B-P2 confirmed DONE):** SPA is fully **en+kn+hi (DONE)**; a
real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical engine bug fixed** (#281). **?
NEXT BUILD (owner-picked): (1) Path B ? point at a bucket video IN THE UI.** ? **B-P1 VERIFIED
ALREADY DONE ? NO ENGINE CODE (do NOT re-scope it as a feature):** `POST /api/process` has
accepted `gs:///`/`gcs:///` URIs since **M2/#280** (submit resolves the URI + skips the local-
exists-404 for non-local backends `main.py:283`; the worker fetches to scratch
`acquire_local_input main.py:159` ? same hash?validate?segment?DB pipeline ? queryable API-DB
`video_id`); **TESTED**
(`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` + hosted-
mode-reject / unknown-scheme-400 / CLI variants). Confirmed by a **3-skeptic verification
workflow (0 refuted, gaps:[])** **Scope pin:** holds in DEFAULT single-user mode (hosted-mode
`allowed_video_root` correctly **400s** a bucket URI ? by design). Stale "local-path-only / net-
new" docs **CORRECTED ? khelsutra [#55]** (DRY_RUN_CUJ $1/2/4/6/8 + LOCAL_APPLIANCE + README).
**? So Path B = ONLY B-P2 = the SPA ingest/progress screen** (enter a `gs:///` URI ? `POST
/api/process` ? poll `/api/jobs` ? load `result.video_id`; back-half already works) ? **100%
khelsutra `web/` SPA + i18n (en/kn/hi), ZERO engine code. **Ready B-P2 sketch: `DRY_RUN_CUJ.md`
$2 (new `web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` +
`App.tsx` videoId?state handoff + `ingest.*` keys). See [[lesson-verify-engine-surface-before-
scoping]]. ** (2) Droplet deploy (opt 3):** deploy the FIXED engine to `168.144.126.176` (un-
provisioned, 2-core/3.8GB) ? re-run CUJ on Linux. **STILL OPEN (localization DoD):** B3 (`site/`
marketing still English-only) + native-speaker review of the AI-drafted kn/hi. **Ramp-up:**
`khelsutra/docs/DRY_RUN_CUJ.md` + `FRIENDLY_AND_MULTILINGUAL.md` $7 + the dry-run checkpoint
below. **Clean slate (at session start):** both repos on master, 0 open PRs, no worktrees/VMS.
khelsutra master `2421557` · engine master `2afd54f` (? origin/master `cee4baa`, #283 M4
squashed).
19
20 **Checkpoint:** 2026-06-21 (ENGINE SERVING M5+ ? OWNER BATCH-AUTHORIZED the independent
path). Owner redirected: "first surface the owner-gated decisions so I can take them + unblock
max independent PRs." Ran a 6-agent grounded gate-map **workflow** (`wf_4849400f`) over README
$2/$7/$8 + code ? a decision menu splitting every M5?M12 gate into ** (a) default-OFF code I can
merge now** vs ** (b) thin owner residual**. **Owner answers (AskUserQuestion):** ** (1) BLANKET
YES** ? write+merge ALL default-OFF serving code (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each; **+ SPEND OK up to $100/?10K** for due-diligence +
**production-grade testing/verification** (so the real M6/M7 cloud runs are ALSO unblocked
within the cap ? must respect it, DELETE resources after, confirm cmd+cost before each spend per
[[gcp-vm-always-delete]]). ** (2) MERGE each on green CI** (OAuth real-testing deferred to a
later rigorous pass). ** (3) ALL recommended design picks** "with generous budget in mind": **co-
located stitch** (D4) · **libx264 ENCODE + NVDEC DECODE-only** (protect parity) · **hard-reject
`<2GB` v1** (config knob) · confirm `COMPUTE_DECOUPLED_SERVING/` name · **M6 = Cloud Run JOBS +
bucket-poll completion**. **? PLANNED PR ORDER:** A(doc: $8 resolutions + P3 reconcile + tracker
housekeeping) ? B(M5 preset+startup-checks+L1?L4 parity test) ? C(M6 `backend/compute/` registry
+ Dockerfile + mocked-dispatch) ? D(M8 v6 migration + pricebook[$0 placeholder] + USD/INR
aggregation + /cost API) ? E(M9-auth Cf-Access JWT verify + tenant scoping; M9-consent inert
NULL cols EXTEND D's v6 migration) ? F(M11 capture?store?shadow-eval?goldenize plumbing,
`flywheel.*`=OFF) ? G(M12 `GDriveApiStorage`+`gdrive-api:///`, fake-Drive tests) + Ops-Agent
codify into `deploy/gcp/provision.sh`. Coordinate: D/E/M9-consent share ONE v6 migration (author
in D, extend in E). **Genuinely owner-only residual:** real GPU run (M5 runlog) [or I run on
cloud GPU within cap], counsel ToS/DPA (M9-consent text), Google OAuth app (M12 real), real GCP
prices+FX (M8 calibrate), CF team-domain/AUD+ingress-lock, repo rename, D0 droplet/token retire.
? shared checkout ? sibling merged **#284** (`06ff4b7`, bucket-URI regression guard) mid/just-
before this session; RE-FETCH origin/master before each branch. **? PROGRESS (this session):** ?
**PR-A [#285] MERGED** (`fefa412`, doc: $8 D16/D17 resolutions + P3 reconcile + tracker
housekeeping). ? **M5 [#286] MERGED** (`b5af66e`, all 7 CI green) ? `backend/config/startup.py`
per-profile fail/warn-fast checks (cloud-serving+non-cloud-storage = the one fatal coherence

```

```

error; GPU/creds = warnings) wired into server (`_enforce_startup_or_exit`?exit1) + worker
(`cmd_infer`/`cmd_train`?rc2), **default-OFF + probe gated behind active profile (worker stays
torch-free at profile='')**; **L4 profile-parity test** (`tests/test_profile_parity.py`: same
stub trajectory?byte-identical windows+ranges across truly-local/cpu-mock/no-profile + worker-
vs-in-process); 4-lens adversarial review folded (`wf_34671ac1`: MAJOR torch-free fix + minors).
Runlog template pre-staged; the real-GPU truly-local **runlog = owner L5**. M5 ? in $7 tracker.
? **M6 [#287] MERGED** (`8cdf65c`, all 7 CI green) ? `backend/compute/` ComputeBackend registry:
`get_compute_backend(config)` returns a backend ONLY for `compute_target=cloud_run` (else `None`
? worker's in-process path, **default-OFF**); `CloudRunCompute` dispatches a Cloud Run **Job**
(injectable `CloudRunJobClient`; real `GoogleCloudRunJobClient` lazy google-cloud-run) ?
**bucket-polls** a done-marker via `execution_policy.poll_until`+storage (D16); container forced
**inline** (`compute_target=local_gpu`+native, never re-dispatch); worker `--done-uri` writes a
success marker. `deploy/cloudrun/{Dockerfile,README}` = L4 image (CUDA+ffmpeg+DejaVu
fonts+wasb` conda torch-1.11 env mirroring gpu_setup.sh; weights staged per WASB-C3) + M7
runbook. 4-lens review (`wf_87b22fa4`) folded a **BLOCKER** (cloud `fetch` needs a dst ? reads
silently returned `{}` on gcs/s3; tests passed only b/c local) + poll-wait test gap +
`PolicyTimeout`?clean rc 4. Suite 1167 green. ? **M8 [#288] MERGED** (`43eccc1`, 7 CI green) ?
`backend/billing.py` (pure `usage`+pricebook, USD source-of-truth / INR display) + **v6
migration** (NULLABLE cost cols on `jobs` + versioned `pricebook` seeded `placeholder-v0` at $0;
additive-NONE, idempotent) + `record_job_cost`/`get_job_cost`/`get_video_cost` + `GET
/api/jobs|videos/{id}/cost` + `BillingConfig` (**default-OFF**) + gated
`_maybe_record_job_cost`. 3-lens review (`wf_498b24af`) folded a **MAJOR silent-GPU-under-bill**
(an unpriced `gpu_type` is now SURFACED ? WARNING + `unpriced_gpu_seconds` breakdown line, not
invisible $0) + the **honest no-op** caveat (pipeline doesn't emit `result['usage']` yet ? cost
0 until a follow-up wires the metering; documented + tested) + margin/rounding reconciliation.
Suite 1186 green. ***? 4 SERVING PRs THIS SESSION, ALL MERGED: #285(docs $8 D16/D17 + P3
reconcile) + #286(M5 startup-checks+L4 parity) + #287(M6 Cloud Run dispatch shim+image) +
#288(M8 cost ledger).** Each default-OFF, isolated, adversarial-reviewed, full-suite+CI green,
tracker+changelog in-PR. ***? v6 is now SHIPPED (M8 took it) ? the NEXT migration is v7**
(M9-consent cols + per-video P1 `workspace_uri` ? the earlier "extend v6" plan shifts to a fresh
v7 `_migrate_to_v7`). ***? REMAINING authorized batch (ready to build next, blueprints loaded):**
**M9-auth** = add `PyJWT[crypto]`; a FastAPI dep extracts `Cf-Access-Jwt-Assertion` ?
fetch+cache the team JWKS ? verify sig/exp/iss/aud ? `user_id`; per-tenant scoping on the v4
`user_id` cols; tighten the placeholder CORS (`main.py:40`); all behind
`security.edge_auth_enabled=False`; offline RSA/fake-JWKS tests. **M9-consent** = inert NULL
`consent_status/is_adult_attested/retention_class` in a **v7** migration (safe no-train default)
? the ToS/DPA TEXT is owner/counsel. **M11** = `flywheel.capture/shadow_eval/promote_stub`
reusing `workspace.py::for_video` + `backend/eval/experiment.compare`(:485) +
`promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked `consent_attested`.
**M12** = `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme against an INJECTED fake
Drive (mirror `gcs.py` DI), keep the mount-only `gdrive://` (truly-local) untouched. **Ops-Agent
codify** into `deploy/gcp/provision.sh` (logging+monitoring APIs + SA metric/log-writer roles +
startup-script ? [[gcp-ops-agent-enable]]). ***? owner-only residual:** real GPU run ? M5 truly-
local runlog (template pre-staged in `runlogs/`); GCP spend for the real M6/M7 run within the
**$100/?10k cap** (build the image per `deploy/cloudrun/README`, run on L4, capture
DEPLOY_REPORT, **DELETE after**); counsel ToS/DPA (M9-consent); real GCP prices+FX + the M8
usage-emission follow-up; Google OAuth app (M12 real); CF team-domain/AUD + ingress-lock; repo
rename; DO droplet/token retire. ***? REAL M7 CLOUD GPU RUN ? ? DONE + VERIFIED (owner:
"complete any and every cloud run with GPUs").** Provisioned an **L4** `rally-m7-gpu`
(`g2-standard-4`, **asia-southeast1-a** Singapore ? both Mumbai zones STOCKOUT for L4; reads the
Mumbai bucket cross-region) ? `bootstrap --gpu` + `gpu_setup.sh` (wasb conda env, **torch
1.11+cu113 cuda True on L4** via PTX-JIT) + staged the 5.82 MiB weights ? `worker -v infer
--video-uri gcs:///mahadevpura_smoke_180s.mp4 --out-uri gcs:/// --segmenter wasb_hybrid` ? **22
rallies** (5395 trajectory points, 2930 visible; Phase 2 WASB 363.7s) ? pushed `gs://khelsutra-
rally-corpus/outputs/05e0e577?/[intervals.csv,report.json]` (segment_count 22, status success).
**VM DELETED, audit 0 instances/0 disks ? $0** (~18 min ? **$0.21**, well under the cap). ***?
SOLVED the long-flagged "cold-start gives 0 rallies" mystery ? THE MISSING PIECE = `--set
indexing.skip_ai_handoff=true`:** `wasb_hybrid` does a Phase-3 **Gemini handoff** by default;
with no `GEMINI_API_KEY` on the box it errors ? 0 segments. `skip_ai_handoff=true` = the
**fully-local** path ? emits the 22 WASB candidate windows directly as rallies (no Gemini). So a
no-Gemini cloud run MUST set `skip_ai_handoff=true` (this is exactly M5's truly-local preset,
which sets it). ? minor: `-v` is a TOP-LEVEL flag ? `worker -v infer` ? (not `infer -v`).
DEPLOY_REPORT runlog = TODO (after M9-auth). ? **M9-auth [#290] MERGED** (`aac7633`) ?
`backend/api/auth.py` Cf-Access JWT verify (RS256: sig+exp+iss+aud; lazy `PyJWT[crypto]`;
injectable JWKS ? fully-offline tests incl. alg=none + HS256-confusion rejection) +

```

```

`resolve_tenant` ? `create_job(owner_id=?)`;
`SecurityConfig.{edge_auth_enabled=False,cf_team_domain,cf_access_aud}` + CORS via
`RALLY_CORS_ALLOW_ORIGINS` env. **Default-OFF.** Security review (`wf_9b4c71d0`) found NO
bypass. ? **OPS-AGENT codify + DRY-RUN GUIDE [#291] MERGED** (`7835f34`, owner asked):
`deploy/gcp/create_gpu_vm.sh` (THE one-command way to spin a GPU VM ? zone-sweep L4
Mumbai?Singapore + **always bakes the Ops-Agent startup-script**) + `ops-agent-startup.sh` +
provision.sh (logging/monitoring APIs + SA metric/log-writer roles ? the empty-graphs root
cause) + **`docs/CLOUD_GPU_DRYRUN_GUIDE.md`** (step-by-step local-video?cloud-GPU?reel,
validated from the M7 run + the local `VideoCompiler` stitch; the **`skip_ai_handoff=true`** no-
Gemini path; `min_shot_count:0` so no rally is filtered). **?? 6 SERVING/INFRA PRs MERGED THIS
SESSION: #285,#286(M5),#287(M6),#288(M8),#290(M9-auth),#291(ops-agent+guide) + the REAL M7 L4
run (22 rallies, $0).** **? NEXT (owner "keep going"): M11** flywheel plumbing
(`flywheel.capture/shadow_eval/promote_stub` reusing `workspace.py::for_video` +
`backend/eval/experiment.compare` + `promotion.py::promote_if_served_safe`, behind
`flywheel.*`=OFF + a faked `consent_attested`) ? **M12**
`GDriveApiStorage(StorageBackend)`+`gdrive-api://` (injected fake-Drive DI; keep mount-only
`gdrive://`). ? M9-consent cols = a **v7** migration (M8 took v6). Ops-Agent codify = ? DONE
(#291; **owner CONFIRMED Observability live in the cloud console 2026-06-21**). ? **#292
MERGED** (`2afbadf`) ? dry-run font fix (`bootstrap.sh` installs `fonts-dejavu-core`; a 2nd dry
run via `create_gpu_vm.sh` PROVED Ops-Agent auto-installs + found the watermark needs the font)
+ the **UI-trigger tracking** (NEXT_STEPS ?P0-0 + [[ui-driven-cloud-serving-gap]]). **??
SESSION WRAPPED ? CLEAN SLATE: 7 PRs merged (#285/286/287/288/290/291/292) + the real M7 L4 run
(22 rallies); 0 open PRs; 0 GPU VMs (total spend ~$0.61); empty m11/m12 placeholder branches
deleted (they had ZERO commits ? the "13k pending lines" the owner saw = untracked
`artifacts/+`scratch/+`sibling docs, NOT my work).** ? **FRESH-SESSION ICEBREAKER = bottom of
[[session-handoff]] $"You are here".** ? Owner's #1 NEXT = make it **UI-DRIVEN + alpha-
shareable** (khelsutra `web/` B-P2 ingest screen + M7 Cloud-Run *service* deploy + M9-on + a
compute picker), THEN M11(flywheel, privacy-sensitive)/M12(gdrive-api). [[gcp-vm-always-delete]]
[[gcp-ops-agent-enable]] [[ui-driven-cloud-serving-gap]] [[working-agreement-merging]]
[[feedback-launch-quality-bar]] [[compute-stack-gcp-only]]
```

21

22 ****Checkpoint:**** 2026-06-21 (STALENESS-FIX / Path-B verify) ? owner: pick up Path B +
"fix all stale docs and stop next sessions spiralling into the staleness." ****? Verified-FIRST
finding: B-P1 was ALREADY DONE**** ? `POST /api/process` accepts `gs:///`gcs:///` since
****M2/#280**** (resolve?skip-exists-404?worker fetch?same pipeline?queryable API-DB `video\_id`) +
TESTED; confirmed by inline file:line trace AND a ****3-skeptic verification workflow (0/3
refuted, gaps:[])****. So the doc-scoped "build the gcs ingest endpoint" was a phantom; Path B =
****SPA-only (B-P2)****. ****DONE this session:**** (1) ****khelsutra [#55] MERGED**** (master `d0461fd`) ?
corrected `DRY\_RUN\_CUJ` \$1/2/4/6/\$8 + `LOCAL\_APPLIANCE` (`/api/health` EXISTS now M1/#268;
bucket-URI ingest NOT a net-new endpoint; T12 engine-half ? done) + `README`; added the anti-
spiral convention ****re-verify engine-surface`[gap]`/`net-new` claims at a cited file:line ? the
engine ships PRs between sessions.**** (2) ****Memory steering corrected**** (this NEXT-SESSION-FOCUS
block + the dry-run checkpoint were stale "local-path-only") + new lesson [[lesson-verify-
engine-surface-before-scoping]] + MEMORY.md index. (3) ****engine [#284] MERGED**** (master
`06ff4b7`, all CI green incl. full suite 2m32s) ? regression GUARD
`test\_process\_bucket\_uri\_yields\_queryable\_video\_id` (parametrized gs://+gcs://) pinning
`video\_id`=MD5(fetched bytes)` (not the URI) + `POST /api/query` queryability (the SPA contract;
previously only LOCAL had it); built+removed in an isolated worktree off `origin/master` (engine
checkout left on its concurrent-session branch). ****? NEXT: B-P2**** = the SPA ingest/progress
screen (enter gs:// URI ? /api/process ? poll /api/jobs ? load video_id; back-half already
works) ? ****100% `web/` + i18n (en/kn/hi), ZERO engine code****; ready sketch in `DRY\_RUN\_CUJ.md`
\$2 + the workflow synthesis
(`web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` + `App.tsx`
videoId?state + `ingest.\*` keys). Then opt-3 droplet (deploy fixed engine + GCS creds + #281 fix
? re-run CUJ on Linux). ****? Repos SYNCED TO HEAD at session end:**** khelsutra master `d0461fd`;
****engine**** working tree `06ff4b7` ? NOTE the engine main checkout couldn't switch to `master`
(sibling worktree `tier1-windowing` holds it), so it's the merged-but-remote-gone
`feat/serving-m4-device-context` branch ****reset --hard to origin/master + upstream repointed to
origin/master**** (safe: tracked tree clean + the 2 divergent commits were the M4 #283 squash,
byte-identical to master; old tip tagged `\_pre\_sync\_m4\_44c9f01` + reflog). So `git pull --ff-
only` works there now; the branch NAME is cosmetic (points at master head).

23 - ****? B-P2 BUILT (in-UI bucket ingest) ? khelsutra
[#56](https://github.com/avidullu/khelsutra/pull/56) OPEN, CI GREEN (web+site+CF build),
awaiting owner merge**** (feature PR ? left for owner; precedent exists for me merging SPA PRs on
owner's "merge as you go"). 100% `web/` + i18n, ZERO engine code (rides B-P1). New `web/src/{api

/types,api/client(processVideo/getJob),lib/errors(parseDetail+classifyError),hooks/useIngestJob(idle?submitting?polling?done|error; recursive-setTimeout poll backoff+10min-ceiling+1 cid; success-shaped-failure?emptyResult),components/IngestPanel}` + `App.tsx` videoId useMemo?useState + IngestPanel-when-!videoId + history.replaceState; `ingest.*` en/kn/hi (AI-draft, native-review-pending). \*\*123 web tests (+19), tsc/build green, browser-verified\*\* (panel renders en+hi; live submit?friendly localized "is the engine running?"; cid logs). Live-fixed UX: bare 5xx/proxy-down ? friendly `networkError`, not "HTTP 500". \*\*? ALL DONE + MERGED (owner: "merge green PRs, improve coverage, leave a CUJ screencast in the promo bucket; proof-and-validation-driven"):\*\* \*\*56 MERGED\*\* (`4525c5d`). \*\*LIVE e2e CUJ recorded\*\* ? real local engine (Gemini + `google-cloud-storage`+ADC for the gs:// fetch), Playwright-driven: SPA ingest `gs://mahadevpura_smoke_180s.mp4` ? \*\*12 rallies\*\* ? Build ? \*\*reel downloaded (1080p, 76s)\*\*. \*\*? Recording it SURFACED A REAL BUG:\*\* `/api/compile` 500'd on a bucket-ingested video ? `videos.filepath` keeps the `gs://` source URI (M2 provenance), handed straight to the stitcher ? ffmpeg can't open `gs://`. \*\*FIXED ? engine [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289) MERGED\*\* (`e68b583`): compile resolves the source via `storage.acquire_local_input` first (identity local / fetch-to-scratch bucket) + regression test; full suite \*\*1182?\*\*. \*\*Screencast ? `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`\*\* (42s, fast-forwarded Gemini wait + frozen reel-ready climax). \*\*DoD \$6 box3 ? ? khelsutra [#57] MERGED\*\* (`7750afc`, +\$9 run result). Engine+SPA+worktrees cleaned up. ? `google-cloud-storage` now pip-installed in the local engine env (enables local gs:// fetch). \*\*? DOC-SCAN + PLAN (2026-06-21, owner-directed):\*\* Ran 2 scans ? (a) \*\*Cloud-Run/serving\*\* is the CONCURRENT session's active line (COMPUTE\_DECOUPLED\_SERVING; M1-M8 merged, M9 #290 open; `backend/compute/cloud_run.py` shim + `deploy/cloudrun/Dockerfile`; \*\*don't duplicate\*\*); key gap = API `/api/process` NOT yet wired to the ComputeBackend seam, only `backend.worker infer` is); (b) **all-docs next-steps** ? prioritized roadmap. **Owner clarification:** "trigger from UI for alpha users" is NOT M12 + NOT GPU-gated ? it = a **hosted SPA+engine behind edge auth on a reachable box** ; cheapest = **the parked CPU droplet + Gemini (CPU, no GPU sub needed)** . **Droplet was parked only for the WASB/GPU CUJ; Gemini alpha doesn't need GPU.** **? EXECUTION PLAN (owner-approved, merge-on-green):** doc-hygiene ? ? **B3** (site/ shared `t()` shim ? NEXT) ? **A1** (plain-lang copy) ? **A4** (mobile touch) ? **A5** (friendly error map) ? **A8** (bg-processing chips) ? **NORTH STAR: runnable tool for alpha = T4 (in-UI file UPLOAD ? keystone; chunked-upload endpoints exist, SPA must call them) ? T2 (droplet appliance shell: serve SPA + reverse-proxy + Cloudflare Access + Gemini; rotate key) ? T11 (retention)** . **A2 (WhatsApp/social share) DROPPED** ? owner defers all social/sharing to a separate scope; reel stays pure-download (wa.me can't attach files; real file-share = Web Share API = own scope). **DONE THIS SESSION (post-CUJ):** **[#58](https://github.com/avidullu/khelsutra/pull/58) MULTILINGUAL_EXPANSION proposal OPEN ? owner will merge** (add th/id/da/es/zh-Hans/ja/ko, en/kn/hi-rigor; researched CLDR plurals [es=3-form {one,many,other}!] + OFL Noto fonts [CJK=tens-of-MB offline] + zh-Hans-vs-Hant; phases A Latin?B Thai?C CJK; **do NOT start until owner approves E0**); **[#59](https://github.com/avidullu/khelsutra/pull/59) doc-hygiene MERGED** (`a0475bc` ? README DRY_RUN delivered + #58 ptr; LOCAL_APPLIANCE T12 ?; REQUEST_ACCESS P2 ?; PER_RALLY live-CUJ note). ? engine-side doc staleness (DOC_STATUS/NEXT_STEPS lag serving M5-M8; RALLY_QUALITY n=6-vs-15) LEFT for the serving session (avoid collision). [[gotcha-shared-checkout-branch-collisions]] [[working-agreement-merging]]

24

25 **Checkpoint:** 2026-06-21 (NEW SESSION ? UX+l10n focus STARTED) ? ramped up via a research+verification **workflow** (web/+site/ string maps, Indic-l10n prior art, **5 adversarial verdicts:** both surfaces **ZERO i18n today** ; recording-edu **already shipped** on the homepage gist + `capture`; `I18N\_PLAN` targets the **dead** `backend/api/static` so its P1/P3 wiring retargets to site/+web/). **? Isolated the work into ONE umbrella tracked project ? `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`** (A: non-tech UX + B: kn/hi l10n; \$7 P-tracker = source of truth; **archive when en+kn+hi render on BOTH surfaces**). Pointers wired: khelsutra `docs/README`+`NEXT\_STEPS`; engine `I18N\_PLAN`+`UI\_PROPOSAL` banners + `DOC\_STATUS` \$3a/\$3b rows (engine edits in an **isolated worktree off origin/master** ? engine checkout is a concurrent session's branch; their findings-#1/#2 PR **274 merged mid-session** ? master `03fa951`). **Owner (AskUserQuestion) locked D6 (en+kn?hi) · D7 (OFL Noto self-host) · D8 (SPA-first) · D9 (catalogs in khelsutra); only D-BACKEND-DEPTH deferred. 3 PRs OPEN:** khelsutra docs [#41](https://github.com/avidullu/khelsutra/pull/41); engine pointers [#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275); **B1 SPA i18n scaffold + language switcher (en-only) ? [#42](https://github.com/avidullu/khelsutra/pull/42) ? CI GREEN (web+site pass), verified LIVE in-browser** (switcher EN|???? persists; content falls back to en with no kn catalog; active-highlight follows the *chosen* locale ? a bug caught by browser verification + fixed). B1 = react-i18next + ICU + detector + `en` catalog + the SPA's FIRST jsdom render tests (79 tests green); deps MIT/BSD-clean. Engine pointer PR built in an isolated worktree (now removed); engine checkout kept flipping branches (concurrent session

active) ? never disturbed. **??** NEXT slices (tracker \$7): B2** (full ~50-string extraction + key-parity/no-bare-strings CI) ? ****B3**** (site `t()` shim) ? ****B4/B5**** (Noto fonts + kn catalog via Gemini-draft?review). ****Also** (owner ask, 2026-06-21): the en/kn/hi academy flyers already existed** (a sibling merged PR ****#40**** ? master `37053b9`; flyer.html / flyer-kn.html / flyer-hi.html served at /flyer{, -kn, -hi}) but were UNLINKED ? ****linked** all three from the homepage footer + added VM-host clean routes ? [#43](https://github.com/avidullu/khelsutra/pull/43) (MERGED ? master `f121c7f`), ****** Then ****owner** asked for MORE flyer placement ? [#44](https://github.com/avidullu/khelsutra/pull/44): "Flyer" in the desktop top-nav + the 3-language links on own-model's foot** (44 site tests green; verified live). (Flyers use Google-Fonts Noto ? separate from the app's D7 self-host; not refactored.) ****Owner** #41 comment ? added a rigorous \$6.5 "automatic language-negotiation" testing strategy** (matrix N1?N9: a kn/hi BROWSER auto-gets the respective version; DoD gate) **+++** concrete browser-language detection tests in B1 [#42]** (N1/N3?N7; 84 web tests green). #41/#42/#275/#44 all merged. **???** THEN owner: "take it to completion until you can run a screencast with a kannada-default browser ? right page renders ? switch hi ? en; merge PRs as you go." ? ? DONE for the `web/` SPA.**** Merged** in sequence (I merged each on green CI, owner-authorized): ****B2** [#45]** (full ~50-string extraction + key-parity/no-bare-strings CI) . ****B4** [#46]** (self-host OFL Noto Kannada/Devanagari, @fontsource-variable) . ****B5** [#47]** (Kannada catalog) . ****B6** [#48]** (Hindi catalog ? the extensibility proof: just a catalog + supported-list entry). ****SPA** now fully en+kn+hi; 99 web tests green** (incl. N1/N2 browser-detection, B5/B6 render, key-parity kn==hi==en, `` sync). **???** SCREENCAST PRODUCED** (Playwright `locale: kn-IN` ? Kannada auto-renders ? switch hi ? en; `C:\Users\avidu\Projects\khelsutra-screencast\khelsutra-i18n-screencast.mp4` (9s) + per-locale frames). Tracker milestone ? [#49]. ? ****kn/hi** are AI-DRAFTED (me) ? native-speaker review pending** before wide public launch. **???** REMAINING for full \$9 DoD (NOT archive-ready):**** **B3**** (the `site/` marketing surface ? homepage/capture still English-only ? via the shared `t()` shim) . ****A3**** recording-education kn/hi . the ****non-tech UX A-items**** (A1/A2/A4/A6...) . native review. Playwright installed in isolated `C:\Users\avidu\Projects\khelsutra-screencast\` (kept out of web/ deps). ****THEN** owner polish batch (2026-06-21) ? all merged:**** (1) ****OFFLINE/self-contained verified + guarded ? [#50]:**** the SPA** is air-gap-renderable (Noto = local woff2 via @fontsource, `@font-face` url?local `assets`, no Google-Fonts/CDN, `web/index.html` no external link); CI guard added so it can't regress. *(The marketing `site/` still uses Google-Fonts ? hosted/online surface, not the offline tool; self-host = optional follow-up.)* (2) ****SCREENCAST** uploaded ? `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`** (+ local). (3) ****Localized SAMPLE-QUERY PROMPT** (en/kn/hi) ? [#51]:**** `query.hint`** = instructive "type `over 10s` for rallies >10s ? or tap a sample" + `query.try` labels pills "Sample queries"; verified live kn. (4) ****\$6.5 testing rule:**** i18n testing is now a MANDATORY merge-gate for any future query-parsing/semantic-understanding/NL work (parser is English-grammar-only today, A9). ****103** web tests green.**** NO** open PRs. [[next-session-ux-localization]] [[doc-status-register]] [[working-agreement-tracked-project-docs]]

26

27 ****Checkpoint:**** 2026-06-21 (DRY-RUN CUJ + a CRITICAL ENGINE FIX) ? owner: "stitch the UI to the tool; dry-run a ****GCS-bucket video**** via the UI ? ****reel** downloaded local**; repeat in Hindi. Run locally (opt 1) then droplet (opt 3); report both; identify fixes + merge." Chose ****A-now-then-B****, ****Gemini****. ****Tracked goal** ? `khelsutra/docs/DRY\_RUN\_CUJ.md`** ([#53]/[#54]). **???** PATH-A LOCAL RUN DONE (en + hi):**** isolated engine off origin/master** (fresh DB/output worktree) + **`.env`** Gemini key ? **gsutil cp gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4`** ? **POST /api/process {gemini}`** ? ****12 rallies**** (`video\_id 05e0e577?`) ? SPA **npm run dev`** (Vite proxies /api?:8000) **video=05e0e577?`** ? loaded 12 real rallies ? ****Build** ? /api/compile ? reel (1080p, 76.5s) DOWNLOADED to `~/Downloads/khelsutra-dryrun/highlights\_local\_{en,hi}.mp4`** (50.9MB). ****Hindi repeat **** (page/cards/"??? ?????"/"???? ???? ???? ??" all Hindi; reel downloaded). **???** CRITICAL BUG FOUND + FIXED ? engine [#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281) MERGED:**** running** the documented **python -m backend.main --server`** runs main.py as **\_\_main\_\_`** (sets **__main__.db_instance`**), but **job\_worker`/uploads`** read **import backend.main`** ? a SEPARATE module whose **db_instance`** stays None ? the **async-job drain** crashes silently on **None.claim\_due\_job()`** ? ****every /api/process job** hangs in **queued` forever\*\***. Latent: unit tests monkeypatch the executor inline; the UI was verified vs a STUB engine. Fix = dispatch **\_\_main\_\_`?`backend.main`;** + **tests/test_server_entrypoint.py`** (real subprocess guard, FAILS-without/PASSES-with). Full engine suite green (2m23s). ****Unblocked** the run via the single-module invocation** (**python -c "...; from backend import main as m; m.main()"**). ? ****OPT-3 DROPLET** (168.144.126.176) NOT YET RUN:**** reachable** (Linux, py3.12, ffmpeg, 2-core/3.8GB) but ****un-provisioned**** (no `~/rally` engine checkout) ? needs a from-scratch engine install OR a proper deploy of the FIXED engine (the fix applies there too). **???** NEXT:**** opt-3 droplet run** (deploy fixed engine ? re-run CUJ); Path B (in-UI bucket ingest). ? ****CORRECTION** (next-session verify):

the "/api/process is local-path-only / net-new gcs ingest" note here was STALE** ?
`/api/process` already accepts `gs://` (M2/#280, tested); Path B is **SPA-only (B-P2)**, no engine endpoint. (The "worker writes CSV to the bucket not the API DB" bit is the *separate* `worker infer` CLI, not the API path ? that's what the stale claim conflated.) Docs fixed ?
khelsutra [#55]; see the NEXT SESSION FOCUS block above + [[lesson-verify-engine-surface-before-scoping]]. [[monetization-plan]]

28
29 **? HANDOFF (2026-06-20) ? KHELSUTRA.GURU brand site: shipped, live, portable. Resume the product threads below.** Separate track from the indexer rally-detection handoff that follows. Full durable record + ramp-up: [[khelsutra-domain-launch]]. Repo =
`avidullu/khelsutra` (NOT this indexer repo).
30 - **? START HERE (2026-06-20, latest ? big session, ALL MERGED, NO open PRs).** Per-rally **MVP/M0 COMPLETE** + a full product/promo wave. **khelsutra master `a7214ad`** . **engine (Khelsutra/badminton-highlight-indexer) master has M1 [#268]**.

96. assistant (2026-06-21T17:27:55.402Z)

All CI green and coverage verified. Let me record the checkpoint and dismiss the now-completed T11 chip:

97. user (2026-06-21T17:28:04.520Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

98. user (2026-06-21T17:28:08.406Z)

No pending task with id task_2480b9e2. Task ids are not persisted across app restarts, so a chip from before a restart can no longer be withdrawn. Nothing was changed.

99. assistant (2026-06-21T17:28:27.088Z)

The chip predates an app restart so it couldn't be programmatically withdrawn (harmless ? I've recorded T11 as done in the checkpoint).

Done ? PR opened, all-green, coverage verified

[Khelsutra/badminton-highlight-indexer#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) ? `feat(serving): T11 ? retention sweep for stale uploads + caches`

CI: all 11 checks green ?

Ruff · Mypy · Full test suite (offline, mocked) · Golden fixtures cross-OS (ubuntu/macOS/windows) · Built-wheel smoke · Dependency license scan · Syntax+import smoke. (Notably, **Ruff passed in CI** ? confirming the untracked `artifacts/` lint errors I flagged are a non-issue since they're uncommitted.)

Coverage ? no regression (verified empirically, not just asserted)

I measured the base branch in an **isolated worktree at `origin/master`** and compared:

	measured statements	missed	coverage
--- --- --- ---			
`origin/master` (baseline)	9748	1456	85.06%
`fix/t11-retention-sweep`	9748	**1433**	**85.30%**

- **Same 9748 measured statements** ? no `backend/`/`training/` source was touched (proven structurally too: the diff is **only** `tools/` + `tests/`, and `tools/cleanup\_caches.py` imports nothing from the measured scope).
- **Branch missed 1433 ? baseline 1456** ? no line lost coverage; it's equal-or-higher.
- The +0.24% is an environment artifact (the `obsreport` reporting tests ran in my local lane but skip in the worktree/CI lane ? same as why CI's floor is calibrated at 70). The canonical CI lane enforces `--cov-fail-under=70` and **passed on the branch**.

Net: ****coverage did not regress.**** Worktree cleaned up afterward.

Notes

- ****One PR****, as scoped ? T11 is a single engine work item. (There's no second repo to PR here; the khelsutra-side §9 T11 doc lives in the sibling repo, not this checkout.)
- I ****did not merge**** ? you asked me to open the PR and check coverage, and a merge is a separate, hard-to-reverse call. CI is green and the window is owner-confirmed, so say the word and I'll squash-merge it; otherwise it's ready for your review.

100. user (2026-06-21T17:29:00.825Z)

please merge it and based on the first prompt does this complete your task?

101. assistant (2026-06-21T17:29:50.328Z)

I'll merge it (CI green, owner-confirmed window ? and you've now explicitly authorized the merge). First a final mergeable/CI check, then squash-merge:

102. user (2026-06-21T17:29:52.949Z)

```
{"checks":[{"Ruff lint":"SUCCESS"}, {"Mypy (lenient baseline)": "SUCCESS"}, {"Syntax sweep + import smoke (no GPU stack)": "SUCCESS"}, {"Full test suite (offline, mocked)": "SUCCESS"}, {"Golden fixtures cross-OS (ubuntu-latest)": "SUCCESS"}, {"Golden fixtures cross-OS (windows-latest)": "SUCCESS"}, {"Golden fixtures cross-OS (macos-latest)": "SUCCESS"}, {"Dependency license scan (no copleft in runtime deps)": "SUCCESS"}, {"Built wheel smoke (install + launch)": "SUCCESS"}], "mergeStateStatus": "CLEAN", "mergeable": "MERGEABLE", "state": "OPEN"}
```

103. assistant (2026-06-21T17:30:05.255Z)

All checks SUCCESS, `MERGEABLE` / `CLEAN`. Squash-merging:

104. user (2026-06-21T17:30:12.844Z)

```
=== PR state after merge ===
{"mergedAt":"2026-06-21T17:30:09Z","state":"MERGED"}
=== master tip now ===
From https://github.com/Khelsutra/badminton-highlight-indexer
   cb739b90..cb7394b  master      -> origin/master
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
```

105. assistant (2026-06-21T17:30:18.406Z)

Merged into `origin/master` as `cb7394b`. Now restoring the shared checkout to where the concurrent serving session left it (session-start branch) and cleaning up the feature branches:

106. user (2026-06-21T17:30:23.720Z)

```
Your branch is up to date with 'origin/serving/exposure-safety-preflight'.
Switched to branch 'serving/exposure-safety-preflight'
Deleted branch fix/t11-retention-sweep (was 7940355).
error: unable to delete 'fix/t11-retention-sweep': remote ref does not exist
error: failed to push some refs to 'https://github.com/Khelsutra/badminton-highlight-indexer.git'
=== final state ===
on branch: serving/exposure-safety-preflight
tree clean? (only untracked expected):
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
?? docs/PACKAGING_PLAN.md
remote feature branch gone? (empty=gone):
```

107. assistant (2026-06-21T17:31:11.802Z)

Merged and cleaned up. `origin/master` is now at `cb7394b` with T11; the shared checkout is restored to `serving/exposure-safety-preflight` (where the concurrent session left it), and both the local and remote `fix/t11-retention-sweep` branches are gone (GitHub auto-deleted the remote head on merge). Working tree is clean apart from the pre-existing untracked files.

Does this complete the first-prompt task?

Yes ? the engine-side T11 retention sweep is implemented, verified, and merged. Mapping each requirement from the original prompt:

```
| Requirement | Status |
|---|---|
| Retention sweep for `output/` and `uploads/` | ? |
| Purge past a retention window: uploaded source videos + orphan .partial-* | ? 7d window
|
| Purge regenerable `output/` artifacts (frames/staged/proxies) | ? (already did; uploads were the gap) |
| Don't hand-roll ? extend tools/cleanup_caches.py | ? extended, not a new tool |
| Keep DB/CSVs/reels durable per policy | ? |
| Dry-run-first | ? --apply required |
| Loud structured logging | ? window banner + per-category subtotals + per-item PURGED`/`FAILED` + `--json` plan |
| Owner-gated window ? recommend + ask, don't hardcode | ? asked ? 7d / keep reels; conservative dry-run default |
| Branch from `origin/master`, re-verify HEAD, explicit-path staging, ONE PR | ? |
| Full local gate: run_all_tests.py, ruff, mypy, coverage ? floor | ? 1290 pass / clean / clean / 85.30% ? 70, no regression |
| Merged | ? [#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) ? cb7394b |
```

Two honest caveats ? not blockers, but the task isn't 100% "operationally live"

- Tool delivered, not an auto-running job.** The prompt said "a sweep (**tool/job**)."
- The khelsutra-side tracking doc isn't flipped.** `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` §9 (T11) lives in the **sibling khelsutra repo**, not this engine checkout, so I couldn't mark the T11 row done from here. That's a one-line status flip in the other repo.

So: **the engine implementation requested is done and merged.** Want me to take either follow-up ? (a) wire the sweep into a scheduled job for live alpha, or (b) note the doc flip for the khelsutra repo? Both are optional and out of the original PR's scope.

108. user (2026-06-21T17:33:19.842Z)

For 1. please file a new github issue and put all the necessary details in it so that it can be picked up as a task in near future. For 2. yes please do the doc flip via a PR and merge it and notify me when done. this was very quick and well done! Thanks!

109. assistant (2026-06-21T17:33:39.113Z)

Thank you! Let me handle both. First, let me locate the khelsutra checkout (for the doc flip) and check the engine's issue labels (for the issue) in parallel:

110. user (2026-06-21T17:33:41.671Z)

```
=== look for local khelsutra checkout ===
FOUND: C:/Users/avidu/Projects/khelsutra
FOUND: C:/Users/avidu/Projects/Khelsutra
```

--- siblings under Projects ---

Annotation Setup
CODE WITH ERIC
Email-Mining
KhelSutraGuru.jpeg
Vairagi
Verifiers
badminton-highlight-indexer
gmail-attachments-to-gdrive
gtest
khelsutra
khelsutra-b3-wt
khelsutra-screencast
muneem
rally-annotator
sports-data-collector
sports-obsreport

111. user (2026-06-21T17:33:42.868Z)

=== engine repo issue labels ===

bug	Something isn't working	#d73a4a
documentation	Improvements or additions to documentation	#0075ca
duplicate	This issue or pull request already exists	#cfd3d7
enhancement	New feature or request	#a2eeef
good first issue	Good for newcomers	#7057ff
help wanted	Extra attention is needed	#008672
invalid	This doesn't seem right	#e4e669
question	Further information is requested	#d876e3
wontfix	This will not be worked on	#ffffff

112. assistant (2026-06-21T17:34:08.624Z)

Found the khelsutra checkout at `C:/Users/avidu/Projects/khelsutra` (shared with the concurrent UX session ? I'll be careful). Let me file the engine issue (task 1) and check the khelsutra repo state in parallel:

113. user (2026-06-21T17:34:32.633Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/301>

114. user (2026-06-21T17:34:33.173Z)

=== khelsutra remote ===

origin <https://github.com/avidullu/khelsutra.git> (fetch)

origin <https://github.com/avidullu/khelsutra.git> (push)

=== current branch + HEAD ===

master

46e0244 feat(web): A9 ? localized tap-only filter controls (#68)

=== status (short) ===

?? .claude/

?? site/access.db

=== worktrees ===

C:/Users/avidu/Projects/khelsutra 46e0244 [master]

C:/Users/avidu/Projects/khelsutra-b3-wt a2292ef [feat/b3-site-i18n]

Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

115. assistant (2026-06-21T17:34:47.173Z)

Issue filed: ****[#301](<https://github.com/Khelsutra/badminton-highlight-indexer/issues/301>)****.

Now the doc flip. khelsutra main checkout is on `master` (clean); the UX session is working in the separate `khelsutra-b3-wt` worktree. I'll fetch latest and read the T11 §9 section before editing (verify-before-asserting):

116. user (2026-06-21T17:34:51.174Z)

```
=== origin/master tip ===
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
=== locate the appliance doc ===
LOCAL_APPLIANCE_ARCHITECTURE.md
=== T11 / $9 occurrences ===
3:> **Status:** `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5
bridge) . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
5:> **Tracking anchors:** $9 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
25:**Read to get current:** `PER_RALLY_SELECTION.md` ($3/$4/$10 contract anchors), engine
`HOSTING_PLAN.md` (gating), `DECOUPLED_COMPUTE/HANDOFF.md` (what's built),
`VIDEO_LOCALITY_MODEL.md` (privacy/retention).
113:**Reused (no re-plumb):** the full `/api` loop; the in-process worker; the chunked upload
endpoints (un-called); LocalStorage default; the detector seam incl. the landing
`NativeWasbRunner`; the **RallyPicker** design + anchors; the `web/` scaffold; the
`HOSTING_PLAN` gate; the `SETUP_NEW_MACHINE` runbook. **Net-new:** Caddyfile + systemd units;
`GET /api/capabilities`; `GET /api/videos`; `GET /api/reels/{video_id}`; `GET /api/health`; the
SPA console screens; ingest-from-bucket wiring; a retention sweep; (reserved) the bucket-worker
bridge.
130:| Retention | `output/`+`uploads/` originals **never deleted** | sweep / keep-proxy-only
before any non-owner exposure |
150:- **L5 ? Owner hardware e2e (handed off).** Native WASB **byte-parity** on a real GPU box;
real-4K **golden-corpus** runs (Gemini + native); Modern-Standby/`cv2`-vs-`ffmpeg` behavior; a
multi-GB browser upload. Reported back into $9, stated plainly as **not verified** in the
offline suite.
152:## 9. Deliverables & progress tracker ? **source of truth**
168:| T11 | Retention sweep for `output/`+`uploads/` | ? | owner | ? | ? |
169:| T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs
(M2/[#280]) ** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` $9) | T4 | ? | ? | engine #280 + SPA #56 |
178:**Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? auto-delete originals after compile? keep proxies+intervals only? window?
(privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4)
**GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker
worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration
endpoint + how it coordinates with the active worker sessions.
188:- [ ] Retention sweep in place; `GEMINI_API_KEY` rotated; `allowed_video_root` set.
189:- [ ] CI green (L2 contract + L4 regression); coverage ? floor; $9 tracker all ???-resolved.
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

117. assistant (2026-06-21T17:35:14.699Z)

Let me read the exact surrounding context (retention table, \$9 tracker, DoD) so the edits are precise and honest:

118. user (2026-06-21T17:35:15.688Z)

```
123      - **Log-analysis pass before merge (DoD item).** Every PR that adds a code path includes
a check that the path logs meaningfully, and the **logs are inspected during CUJ live-replay**
($8 Layers 173). Ties [[feedback-launch-quality-bar]].
124
125      ## 6. Risk table
126      | Risk | Today | Mitigation |
127      |---|---|---|
128      | No app auth + CORS `` `[main.py:42]` | open if any of {localhost-bind, firewall,
gate} slips | enforce all three; **LIVE posture test** that raw `:8000` is refused from outside
|
129      | Test ports on shared droplet | MinIO `9000/9001` + GCS emu `9023` | firewall closes
them |
130      | Retention | `output/`+`uploads/` originals **never deleted** | sweep / keep-proxy-only
before any non-owner exposure |
131      | Surprise Gemini/GPU bill | **no cost ceiling/ledger** | per-video cap;
```

```
`ai_max_workers` (default 5) tunable; `max_workers=1` job serialization |
132 | Native WASB unverified | partially wired; owner-parity-gated | treat as not-
production-verified until owner byte-parity on a real GPU box |
133 | New engine endpoints | `capabilities`/`videos`/`reels`/`health` don't exist |
coordinate engine PRs with active sessions; **branch from `origin/master`, re-verify HEAD,
explicit `git add`** |
134 | Drive ingest overstated | `gdrive://` is a stub | don't present Drive as live; require
Drive?bucket pre-step until real |
```

119. user (2026-06-21T17:35:22.337Z)

```
152    ## 9. Deliverables & progress tracker ? **source of truth**
153    Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking; CI green gates merge; coverage ? floor; each PR
ships its tests + meaningful logs ($5).
154
155    | ID | Deliverable | Depends on | Gated? | Status | PR |
156    |----|-----|-----|-----|-----|----|
157    | T0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | owner | ? |
(this PR) |
158    | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ?
| ? |
159    | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
160    | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ?
|
161    | T4 | S1 ingest: **local upload wired ?
([#66](https://github.com/avidullu/khelsutra/pull/66))** ? file picker ? chunked `/api/upload/`
? server path ? `/api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
162    | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163    | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164    | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ?
| ? |
165    | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
166    | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
167    | T10 | Observability pass: correlation id end-to-end + structured logs + metrics ($5) |
T4 | no | ? | ? |
168    | T11 | Retention sweep for `output/`+`uploads/` | ? | owner | ? | ? |
169    | T12 | Ingest from **bucket URI** (`s3:///`/`gs:///`) wiring ? engine accepts URIs
(M2/[#280]) ** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs:///`?reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` $9) | T4 | ? | ? | engine #280 + SPA #56 |
170    | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner
byte-parity (P4) | branch | ? owner | ? | ? |
171    | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |
T8,T12 | ? owner | ? | ? |
```

120. assistant (2026-06-21T17:36:18.417Z)

The doc content is current (main checkout `master` == `origin/master` at `46e0244`). I'll make the doc flip in a dedicated worktree off `origin/master` (the established pattern for this shared checkout ? keeps the UX session's tree untouched):

121. user (2026-06-21T17:36:20.753Z)

```
Preparing worktree (new branch 'docs/t11-retention-done')
branch 'docs/t11-retention-done' set up to track 'origin/master'.
HEAD is now at 46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
=== worktree HEAD (must be origin/master 46e0244) ===
docs/t11-retention-done
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```


122. assistant (2026-06-21T17:36:28.676Z)

Now let me read the doc from the worktree path (registers it for editing) and apply the flip:

123. user (2026-06-21T17:36:29.375Z)

```
168      | T11 | Retention sweep for `output/`+`uploads/` | ? | owner | ? | ? |
```

124. assistant (2026-06-21T17:37:00.105Z)

Let me read the exact lines I'll edit (risk table, blocking gate, DoD) from the worktree to match strings precisely:

125. user (2026-06-21T17:37:00.980Z)

```
176
177     ## 10. Open questions ? owner / external
178     **Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? auto-delete originals after compile? keep proxies+intervals only? window?
(privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4)
**GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker
worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration
endpoint + how it coordinates with the active worker sessions.
179     **Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming
endpoint; per-user scoping timing; containerization vs runbook.
180
181     ## 11. Definition of done
182     - [ ] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via
the ingest menu.
183     - [ ] Raw `:8000/`:900x` unreachable from outside (**tested**, not assumed).
184     - [ ] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped
honestly (not faked).
185     - [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
186     - [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via
`/api/videos` + `/api/reels`.
187     - [ ] **Observability:** correlation id threaded end-to-end; structured logs at every
new path; loud failures; logs inspected during CUJ replay.
188     - [ ] Retention sweep in place; `GEMINI_API_KEY` rotated; `allowed_video_root` set.
189     - [ ] CI green (L2 contract + L4 regression); coverage ? floor; $9 tracker all
?/?-resolved.
```

126. assistant (2026-06-21T17:37:04.652Z)

Now I have all exact strings. Applying the four edits ? flip the T11 row, update the risk table, the blocking gate, and the DoD line:

127. user (2026-06-21T17:37:08.747Z)

The file C:\Users\avidu\Projects\khelsutra-t11-wt\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

128. user (2026-06-21T17:37:14.211Z)

The file C:\Users\avidu\Projects\khelsutra-t11-wt\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

129. user (2026-06-21T17:37:20.002Z)

The file C:\Users\avidu\Projects\khelsutra-t11-wt\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

130. user (2026-06-21T17:37:27.862Z)

The file C:\Users\avidu\Projects\khelsutra-t11-wt\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

131. assistant (2026-06-21T17:37:33.447Z)

All four edits applied. Let me verify the diff (only the doc, edits correct), then commit/push:

132. user (2026-06-21T17:37:35.029Z)

=== files changed (must be ONLY the doc) ===

M docs/LOCAL_APPLIANCE_ARCHITECTURE.md

=== diff ===

```
diff --git a/docs/LOCAL_APPLIANCE_ARCHITECTURE.md b/docs/LOCAL_APPLIANCE_ARCHITECTURE.md
index 9cf7840..eefbfdc 100644
--- a/docs/LOCAL_APPLIANCE_ARCHITECTURE.md
+++ b/docs/LOCAL_APPLIANCE_ARCHITECTURE.md
@@ -127,7 +127,7 @@ Launch-readiness directive (owner 2026-06-20): **logs and monitoring taken
serio
|---|---|---|
| No app auth + CORS `` `[main.py:42]` | open if any of {localhost-bind, firewall, gate} slips
| enforce all three; **LIVE posture test** that raw `:8000` is refused from outside |
| Test ports on shared droplet | MinIO `9000/9001` + GCS emu `9023` | firewall closes them |
-| Retention | `output/+`uploads/` originals **never deleted** | sweep / keep-proxy-only before
any non-owner exposure |
+| Retention | sweep **tool** now exists ([engine #299](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/299), dry-run-first, uploads >7d, reels durable) but **not yet auto-
scheduled** ([engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301))
| run the sweep `--apply` before/with any non-owner exposure; auto-schedule per
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301); keep-proxy-only
still an option |
| Surprise Gemini/GPU bill | **no cost ceiling/ledger** | per-video cap; `ai_max_workers`
(default 5) tunable; `max_workers=1` job serialization |
| Native WASB unverified | partially wired; owner-parity-gated | treat as not-production-
verified until owner byte-parity on a real GPU box |
| New engine endpoints | `capabilities`/`videos`/`reels`/`health` don't exist | coordinate
engine PRs with active sessions; **branch from `origin/master`, re-verify HEAD, explicit `git
add`** |
@@ -165,7 +165,7 @@ Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One smal
| T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) | ? |
coord | ? | ? |
| T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no | ? | ?
|
| T10 | Observability pass: correlation id end-to-end + structured logs + metrics (S5) | T4 |
no | ? | ? |
-| T11 | Retention sweep for `output/+`uploads/` | ? | owner | ? | ? |
+| T11 | Retention sweep for `output/+`uploads/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges
stale uploads >7d + orphan `.partial-*` + regenerable frames/staged/proxies; **reels/DB/CSVs
durable**; `--json` plan ready for a scheduler. Automated scheduling ? [engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ?
(uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |
| T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs (M2/[#280])
**+ SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live e2e
(`DRY_RUN_CUJ.md` S9) | T4 | ? | ? | engine #280 + SPA #56 |
| T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner byte-
parity (P4) | branch | ? owner | ? | ? |
| T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) | T8,T12 | ?
owner | ? | ? |
@@ -175,7 +175,7 @@ Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One smal
**? Alpha deploy path chosen = SPLIT, not the single-origin shell (owner 2026-06-21).** The
alpha ships the **split** topology (SPA on Cloudflare Pages + engine on an `api.*` cloudflared
tunnel + CF Access over both ? the engine `HOSTING_PLAN.md` S2), **not** the single-origin Caddy
shell of **T2** (kept as the documented alternative ? S0/D1). The **deploy kit + runbook**
```

```
([`deploy/DEPLOY_ALPHA.md`](../deploy/DEPLOY_ALPHA.md) +
`deploy/cloudflared.config.example.yml`) is shipped (? templates+runbook); the actual stand-up
(tunnel + Pages + CF Access + key rotation) is the **owner** step. The **T3**
raw-:8000-refused posture test is documented as the deploy kit's $6 verification. Enabled by:
engine exposure-safety boot posture (engine #294) + SPA `VITE_API_BASE` ([#64]).
```

```
## 10. Open questions ? owner / external
-**Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? auto-delete originals after compile? keep proxies+intervals only? window?
(privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4)
**GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker
worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration
endpoint + how it coordinates with the active worker sessions.
+**Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:**
auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-
video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU
droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud
Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates
with the active worker sessions.
**Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming endpoint;
per-user scoping timing; containerization vs runbook.
```

```
## 11. Definition of done
@@ -185,7 +185,7 @@ Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One smal
- [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
- [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via `/api/videos`
+ `/api/reels`.
- [ ] **Observability:** correlation id threaded end-to-end; structured logs at every new path;
loud failures; logs inspected during CUJ replay.
-- [ ] Retention sweep in place; `GEMINI_API_KEY` rotated; `allowed_video_root` set.
+- [ ] Retention sweep in place (**tool shipped** [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299); **auto-schedule**
[engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301));
`GEMINI_API_KEY` rotated; `allowed_video_root` set.
- [ ] CI green (L2 contract + L4 regression); coverage ? floor; $9 tracker all ??-resolved.
```

```
## 12. References
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

133. assistant (2026-06-21T17:37:45.591Z)

Clean, honest diff ? only the doc, four targeted edits. Re-verifying HEAD and committing (explicit path):

134. user (2026-06-21T17:37:51.138Z)

```
=== re-verify branch + HEAD (mine, off origin/master) ===
docs/t11-retention-done
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
=== stage explicit path + commit ===
[docs/t11-retention-done 20d1d81] docs(appliance): flip T11 retention sweep to ? ? tool shipped
(engine #299), scheduling tracked (#301)
1 file changed, 4 insertions(+), 4 deletions(-)
=== verify only my commit ahead of origin/master ===
20d1d81 docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

135. assistant (2026-06-21T17:37:58.385Z)

Clean ? only my commit ahead. Pushing and opening the PR:

136. user (2026-06-21T17:38:12.439Z)

```
remote:
branch 'docs/t11-retention-done' set up to track 'origin/docs/t11-retention-done'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      docs/t11-retention-done -> docs/t11-retention-done
=== create PR ===
https://github.com/avidullu/khelsutra/pull/69
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

137. assistant (2026-06-21T17:38:17.012Z)

khelsutra PR **[#69](https://github.com/avidullu/khelsutra/pull/69)** opened. Let me check its CI and merge once green:

138. user (2026-06-21T17:38:24.973Z)

```
{"checks":[{"site":""}, {"web":""}, {"Workers Builds: hidden-king-
khelsutra":""}], "mergeStateStatus": "UNSTABLE", "mergeable": "MERGEABLE", "state": "OPEN"}
```

139. assistant (2026-06-21T17:38:31.434Z)

Checks are running (Cloudflare Pages `web`/`site` builds + Workers). Per the repo's own "CI green gates merge" rule, let me watch them to completion:

140. user (2026-06-21T17:39:04.891Z)

```
web      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351008
Refreshing checks status every 15 seconds. Press Ctrl+C to quit.

Workers Builds: hidden-king-khelsutra   pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/4152ad6f-cdf3-4546-974d-1fea42a7f396
site     pass    10s
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351014
web      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351008
Workers Builds: hidden-king-khelsutra   pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/4152ad6f-cdf3-4546-974d-1fea42a7f396
site     pass    10s
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351014
web      pass    37s
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351008
Workers Builds: hidden-king-khelsutra   pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/4152ad6f-cdf3-4546-974d-1fea42a7f396
site     pass    10s
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351014
web      pass    37s
https://github.com/avidullu/khelsutra/actions/runs/27912275082/job/82591351008
=== watch exit=0 ===
```

141. assistant (2026-06-21T17:39:16.173Z)

All three checks pass. Squash-merging and cleaning up the worktree:

142. user (2026-06-21T17:39:26.158Z)

```
=== PR state ===
{"mergedAt": "2026-06-21T17:39:20Z", "state": "MERGED"}
```

```

=== remove worktree + delete local branch ===
Deleted branch docs/t11-retention-done (was 20d1d81).
=== fetch + confirm master has the flip ===
From https://github.com/avidullu/khelsutra
 46e0244..bea8a2a  master    -> origin/master
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301) (#69)
=== remote branch gone? (empty=gone) ===
20d1d81561c994c05b1e72d308efb3368caf9e92      refs/heads/docs/t11-retention-done
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

143. assistant (2026-06-21T17:39:30.614Z)

Merged into khelsutra master (`bea8a2a`). The remote branch wasn't auto-deleted here ? let me clean it up:

144. user (2026-06-21T17:39:36.525Z)

```

To https://github.com/avidullu/khelsutra.git
- [deleted]      docs/t11-retention-done
=== remote branch gone? (empty=gone) ===
(empty above = deleted)
=== khelsutra main checkout state (left as found: master, clean) ===
master
?? .claude/
?? site/access.db
=== worktrees now ===
C:/Users/avidu/Projects/khelsutra      46e0244 [master]
C:/Users/avidu/Projects/khelsutra-b3-wt  a2292ef [feat/b3-site-i18n]
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

145. assistant (2026-06-21T17:40:00.069Z)

Both tasks complete. Let me update the live checkpoint to the final state, then notify you:

146. user (2026-06-21T17:40:11.765Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

147. user (2026-06-21T17:40:22.131Z)

```

13
14      **Checkpoint:** 2026-06-21 (**T11 retention sweep ? engine PR OPEN, all-green, coverage
verified**). The owner-spun T11 chip (`task_2480b9e2`) is now BUILT in the engine. **Extended
`tools/cleanup_caches.py`** (not hand-rolled, per [[cache-hygiene-policy]]) with a pre-alpha
**upload retention sweep**: a new `uploads` scan location + `classify` (`uploaded source` /
`partial upload` for the orphaned `partial-<id>` files) over `security.upload_dir` (default
`output/uploads`); `plan_deletions` gains an **independent `uploads_retention_days` window**
(privacy) orthogonal to the cache `--older-than` (disk); `scan_windows` now **skips the nested
uploads dir** (was silently kept whole as "other dir" ? no retention); loud structured output
(retention-window banner + per-category subtotals + per-item PURGED/FAILED + a `--json` plan for
a cron/job wrapper). **Dry-run by default; reels/DB/CSVs stay durable.** **Owner gate
(AskUserQuestion): uploads 7d default; reels kept durable.** ? **engine
[#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) OPEN, ALL 11 CI checks
GREEN** (ruff/mypy/full-suite/golden cross-OSx3/wheel-smoke/license-scan/syntax+import); +6 unit
tests; **coverage no-regression VERIFIED** (branch **85.30%** vs `origin/master` baseline
**85.06%** measured in an isolated worktree; same **9748** measured stmts, branch missed **1433
? ** baseline **1456** ? the +0.24% is obsreport reporting tests in my local lane, NOT my diff;
the diff touches **ONLY tools/+tests** and the tool imports **zero** backend/training). Branch
`fix/t11-retention-sweep`, cut from `origin/master` in the shared checkout (serving session's
branch untouched; explicit-path staging). **? Awaiting owner merge decision** (feature-ish; not
auto-merged despite green CI). The khelsutra-side $9 T11 doc lives in the sibling repo (not
editable from this checkout). [[cache-hygiene-policy]] [[ui-driven-cloud-serving-gap]] [[gotcha-
shared-checkout-branch-collisions]]

```

148. user (2026-06-21T17:40:30.335Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

149. user (2026-06-21T17:40:44.578Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

150. assistant (2026-06-21T17:41:03.999Z)

Both done. ?

1. Automation follow-up ? issue filed

****[engine #301 ? "T11 follow-up: run the upload retention sweep as an automated scheduled job (hosted alpha)"](<https://github.com/Khelsutra/badminton-highlight-indexer/issues/301>)**** (labeled `enhancement`).

It's self-contained for a future pickup ? context (the #299 tool + `--json` hook), the gap (no unattended scheduler), what's needed (cron/systemd-timer for single-box, Cloud Scheduler for the Cloud Run path), design considerations I flagged (in-flight `partial` safety, active-job safety via the jobs table, per-user-subdir handling for PR-3, the apply-guard since the scheduled run uses `--apply`), and concrete acceptance criteria.

2. Doc flip ? PR opened, merged

****[khelsutra #69](<https://github.com/avidullu/khelsutra/pull/69>) ? MERGED**** (`bea8a2a`), all CI green (`web`/`site`/Workers builds).

In `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` I flipped ****T11 ? ? ?**** and updated the three inbound references so the doc stays honest rather than overstated:

- ****§9 tracker:**** T11 done, with the engine #299 PR ref + the owner window (uploads 7d, keep reels) + scheduling pointer to #301.
- ****§6 risk table:**** retention row ? "tool exists, not yet auto-scheduled".
- ****§10 blocking gate #2:**** retention window ****DECIDED****; auto-delete-vs-keep-proxy still open.
- ****§11 DoD:**** retention line annotated (tool shipped #299 / auto-schedule #301).

The framing throughout is deliberately honest: the sweep ****tool**** is shipped; ****unattended scheduling**** is the tracked remaining piece (#301) ? not "T11 fully operational."

Hygiene

Doc change was done in a dedicated worktree off `origin/master` (UX session's `khelsutra-b3-wt` left untouched); worktree removed, local + remote `docs/t11-retention-done` branches deleted, khelsutra main checkout left exactly as found (on `master`, clean). Engine `fix/t11-retention-sweep` was likewise cleaned up after the earlier #299 merge.

Net: ****T11** is complete and merged in the engine (#299), the operational follow-up is tracked (#301), and the appliance doc reflects reality (khelsutra #69).******

151. user (2026-06-21T17:42:19.105Z)

amazing! There is one last help if you can, I am trying to crawl <https://khelsutra.guru> in a sibling session and it says its getting 403s, can you help debug it?

152. assistant (2026-06-21T17:42:48.257Z)

Happy to help ? khelsutra.guru is your own site (Cloudflare-hosted, per the deploy setup), so 403s on a crawler almost always trace to Cloudflare's bot protection rather than your app. Let me reproduce it empirically and pin down the exact cause. I'll probe with several user agents and inspect the response headers:

153. user (2026-06-21T17:42:55.486Z)

```
===== status by User-Agent =====
browser      -> HTTP 000
curl-default  -> HTTP 000
python-requests -> HTTP 000
scrapy        -> HTTP 000
googlebot     -> HTTP 000
empty-UA      -> HTTP 000
```

```
===== robots.txt =====
```

```
[HTTP 000]
```

154. assistant (2026-06-21T17:43:09.988Z)

HTTP `000` across the board (even robots.txt) means the request never completed ? that's not a 403, it's my sandbox having no outbound network. Let me confirm, and in parallel inspect your `site/` deploy config (robots.txt / `\_headers` / Worker) which is where an app-level 403 would originate:

155. user (2026-06-21T17:43:12.817Z)

Exit code 43

```
=== verbose curl to confirm network reachability ===
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current	
			Dload	Upload	Total	Spent	Left	Speed
0	0	0	0	0	0	--:--:--	--:--:--	0<!doctype html>

```
<html lang="en">
<head>
<meta charset="utf-8" />
<meta name="viewport" content="width=device-width, initial-scale=1" />
<title>KhelSutra ? every rally indexed, the dead time gone</title>
<meta name="description" content="KhelSutra turns a full badminton match into a clean index of
rallies with the dead time stripped out ? a highlight reel you can watch and download. Invite-
only alpha, Bengaluru." />
<meta property="og:title" content="KhelSutra ? every rally indexed, the dead time gone" />
<meta property="og:description" content="A full match becomes a clean highlight reel ? only the
real play, no dead time. Invite-only alpha for academies and players in Bengaluru." />
<meta property="og:type" content="website" />
<meta property="og:url" content="https://khelsutra.guru" />
<meta property="og:image" content="https://khelsutra.guru/og.png" />
<meta property="og:image:width" content="1200" />
```

```
=== control: can the sandbox reach ANY host? ===
```

```
example.com -> 000
```

156. user (2026-06-21T17:43:14.991Z)

```
=== robots.txt / _headers / _redirects / worker / wrangler in site & repo ===
```

```
site/wrangler.toml
```

```
./site/wrangler.toml
```

```
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

157. assistant (2026-06-21T17:43:44.033Z)

The site **did** respond with the real homepage HTML to a plain `curl` ? so khelsutra.guru does **not** blanket-403 crawlers at the app level. (`example.com` ? 000` confirms my sandbox has flaky/limited egress, so I can't reliably hammer it from here ? but the one success is the useful signal.) The 403 is almost certainly **Cloudflare's edge bot protection**, not your app. Let me confirm there's no app-level 403 logic in your Worker/site config:

158. user (2026-06-21T17:43:44.486Z)

```

1      # Cloudflare Worker for khelsutra.guru ? static site + Request-Access form API.
2      #
3      # One-time setup (owner, our Cloudflare account):
4      #   npx wrangler d1 create khelsutra-access          # copy the printed database_id
below
5      #   npx wrangler d1 execute khelsutra-access --remote --file=schema.sql
6      #   npx wrangler deploy                          # or connect this repo via Workers
Builds (git)
7      #
8      name = "hidden-king-khelsutra" # the git-connected Worker that owns khelsutra.guru ?
keep in sync with Cloudflare
9      main = "src/index.js"
10     compatibility_date = "2025-06-01"
11
12     [assets]
13     directory = "./public"
14     binding = "ASSETS"
15
16     [[d1_databases]]
17     binding = "DB"
18     database_name = "khelsutra-access"
19     database_id = "cd02c6bf-4bd6-4a33-a333-ced86e36b511" # from `wrangler d1 create
khelsutra-access`
20
21     # Email-on-registration: Email Routing send binding -> a VERIFIED destination address.
22     [[send_email]]
23     name = "NOTIFY"
24     destination_address = "avi.dullu@gmail.com"
25
26     [vars]
27     NOTIFY_FROM = "noreply@khelsutra.guru"
28     NOTIFY_TO = "avi.dullu@gmail.com"
29

```

159. assistant (2026-06-21T17:43:50.811Z)

It's a Cloudflare Worker (`hidden-king-khelsutra`) serving static assets + a D1-backed access form. Let me read the Worker source for any 403 logic, and check `public/` for a robots.txt:

160. user (2026-06-21T17:43:50.986Z)

```

1      // Cloudflare Worker for khelsutra.guru ? serves the static site and handles the
2      // Request-Access form: POST /api/request-access -> validate -> insert into D1 (binding
DB).
3      // Pure helpers are exported so they can be unit-tested without a live Worker/D1.
4
5      import { d1Store } from "./store.js";
6      import { cfEmailNotify } from "./notify.js";
7
8      const SEGMENTS = ["academy", "individual"];
9      const GPU = ["yes", "unsure", "no", "prefer-cloud"];
10     const COURTS = ["1", "2-4", "5+"];
11     const STORAGE = ["gdrive", "onedrive", "dropbox", "bucket", "local", "unsure"];
12
13     function str(v, max) {
14         if (v == null) return null;
15         const s = String(v).trim();
16         return s ? s.slice(0, max) : null;
17     }
18
19     // Honeypot: humans never fill the hidden "company" field.
20     export function isBot(p) {
21         return !(p && typeof p.company === "string" && p.company.trim() !== "");
22     }
23

```



```

24     export function validate(p) {
25         if (!p || typeof p !== "object") return { ok: false, error: "bad payload" };
26         const email = String(p.email || "").trim();
27         if (!/^[@\s]+\.[^\s]+$/ .test(email)) return { ok: false, error: "valid email
required" };
28         return { ok: true };
29     }
30
31     export function toRow(p, meta = {}) {
32         const storage = Array.isArray(p.storage)
33             ? p.storage.filter((s) => STORAGE.includes(s)).join(",")
34             : "";
35         return {
36             segment: SEGMENTS.includes(p.segment) ? p.segment : null,
37             name: str(p.name, 200),
38             email: str(p.email, 320),
39             city: str(p.city, 200),
40             org_name: str(p.org_name, 200),
41             courts: COURTS.includes(p.courts) ? p.courts : null,
42             has_gpu: GPU.includes(p.has_gpu) ? p.has_gpu : null,
43             storage,
44             message: str(p.message, 4000),
45             consent: p.consent ? 1 : 0,
46             country: str(meta.country, 8),
47             user_agent: str(meta.userAgent, 400),
48         };
49     }
50
51     function json(obj, status = 200) {
52         return new Response(JSON.stringify(obj), { status, headers: { "content-type":
"application/json" } });
53     }
54
55     export async function handleRequest(request, store, notify) {
56         let body;
57         try {
58             body = await request.json();
59         } catch {
60             return json({ ok: false, error: "invalid json" }, 400);
61         }
62         if (isBot(body)) return json({ ok: true }); // silently drop bots
63         const v = validate(body);
64         if (!v.ok) return json({ ok: false, error: v.error }, 400);
65
66         const r = toRow(body, {
67             country: request.headers.get("cf-ipcountry"),
68             userAgent: request.headers.get("user-agent"),
69         });
70
71         await store.insertAccessRequest(r);
72         try { await notify?.notify(r); } catch { /* email is best-effort; never fail the
request */ }
73
74         return json({ ok: true });
75     }
76
77     export default {
78         async fetch(request, env) {
79             const url = new URL(request.url);
80             if (url.pathname === "/api/request-access") {
81                 if (request.method !== "POST") return json({ ok: false, error: "method not
allowed" }, 405);
82                 return handleRequest(request, d1Store(env.DB), cfEmailNotify(env.NOTIFY,
env.NOTIFY_FROM, env.NOTIFY_TO));
83             }

```

```

84         return env.ASSETS.fetch(request); // everything else: static assets
85     },
86 };
87

```

161. assistant (2026-06-21T17:44:41.623Z)

Found the smoking gun ? by **elimination**. Your Worker ([site/src/index.js](../khelesutra/site/src/index.js)) contains **no 403 anywhere**:

- ``/api/request-access``: non-POST ? **405**, bad/invalid ? **400**, bot-honeypot ? silent **200**
- everything else ? ``env.ASSETS.fetch()`` ? static asset **200/404**

And a plain ``curl`` (default UA, no browser headers) pulled the full homepage. So the 403 is **not your application** ? it's **Cloudflare's edge security** sitting in front of the Worker. Let me confirm there's no robots.txt and check the public assets:

162. user (2026-06-21T17:44:43.889Z)

```

=== public/ contents (robots.txt present?) ===
total 164
drwxr-xr-x 1 avidu 197609      0 Jun 21 13:08 .
drwxr-xr-x 1 avidu 197609      0 Jun 21 12:12 ..
-rw-r--r-- 1 avidu 197609  6983 Jun 20 11:30 capture.html
-rw-r--r-- 1 avidu 197609 13754 Jun 21 13:08 flyer-hi.html
-rw-r--r-- 1 avidu 197609 14410 Jun 21 13:08 flyer-kn.html
-rw-r--r-- 1 avidu 197609 10938 Jun 21 13:08 flyer.html
drwxr-xr-x 1 avidu 197609      0 Jun 21 13:08 img
-rw-r--r-- 1 avidu 197609  43816 Jun 21 12:12 index.html
-rw-r--r-- 1 avidu 197609  40635 Jun 20 11:30 og.png
-rw-r--r-- 1 avidu 197609 15007 Jun 21 12:12 own-model.html
drwxr-xr-x 1 avidu 197609      0 Jun 20 23:27 video

```

```

=== any robots.txt anywhere in site/ ? ===
(empty above = NO robots.txt is served)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

163. assistant (2026-06-21T17:45:50.996Z)

Debugged. Here's the verdict and the fix.

Diagnosis: the 403 is **Cloudflare's edge, not your site**

By elimination:

- Your Worker ([site/src/index.js](../khelesutra/site/src/index.js)) emits **no 403 anywhere** ? ``/api/request-access`` returns 405 (non-POST) / 400 (bad payload) / silent 200 (honeypot); everything else is ``env.ASSETS.fetch()`` ? static **200/404**.
- A plain ``curl`` with the **default UA and zero browser headers** pulled your full homepage HTML ? there's no blanket block, and it's **not Cloudflare Access** (that would return a login interstitial, not your HTML).
- No ``robots.txt`` is served either (that'd be a 404, not a 403).

So the 403 is Cloudflare's security layer in front of the ``hidden-king-khelesutra`` Worker, reacting to **crawler-specific signals** ? bot classification, request rate, or TLS/JS fingerprint ? that a single curl doesn't trip but a crawler does.

Pinpoint which Cloudflare feature (run from the crawler's box ? it has network; my sandbox doesn't)

```

```bash
curl -sS -D - -o /dev/null -A "<your crawler's exact UA>" "https://khelesutra.guru/<the exact path that 403s>"
```

```

Read the status line + these headers ? they tell the three suspects apart:

```
Cause	Tell-tale in the 403
**Bot Fight Mode**	`cf-mitigated: challenge`, or body "Checking your browser"
**WAF rule**	body "Sorry, you have been blocked" + a Ray ID
**Rate limiting**	status **429**, or 403 with a `Retry-After` header
```

Authoritative source: **Cloudflare dashboard ? khelsutra.guru zone ? Security ? Events**, filter Action = Block/Challenge. Each row names the exact service (Bot Fight Mode / WAF / Rate Limiting) and the Ray ID. That's the definitive answer in ~10 seconds.

Fix ? it's your own site, so allowlist your crawler (don't weaken protection globally)

1. **Most reliable, works on the free plan ? IP Access Rule = Allow.** Security ? WAF ? Tools ? IP Access Rules ? add your crawler's **egress IP**, action **Allow**, scope this zone. An "Allow" IP rule bypasses Bot Fight Mode **and** security level. (Pin the crawler's outbound IP first: `curl -s https://api.ipify.org`` from the crawler.)
2. **Pro+ plan ? WAF Skip rule by secret header.** Security ? WAF ? Custom rules ? expression `any(http.request.headers["x-crawl-key"][*] eq "<random-secret>")``, action **Skip ? Super Bot Fight Mode + All managed rules + Rate limiting**; then have the crawler send `X-Crawl-Key: <random-secret>``. Robust regardless of UA/IP. *(Note: plain Bot Fight Mode on the free plan can't be selectively skipped ? there you either use the IP allowlist in #1 or toggle it off.)*
3. **Quick & dirty ? toggle Bot Fight Mode off** (Security ? Bots) for the crawl window.
4. **If it's rate limiting:** lower concurrency + add a delay (~1 req/s). A burst of identical-fingerprint requests trips both BFM heuristics and rate rules even with a perfect UA.
5. **Crawler-side, no CF change:** send a real browser UA + `Accept`` + `Accept-Language``. Light crawling may pass (curl did), but BFM fingerprints beyond UA, so #1/#2 are the durable fixes.

My pick: **#1 (IP allowlist)** if your crawler has a stable egress IP ? one dashboard entry, no app change, works on any plan.

Want me to land two quick PRs while I'm here? (a) add `public/robots.txt`` (declare crawl rules + sitemap; good hygiene since you're crawling), and (b) a short "crawling your own site" runbook in the site deploy docs with the IP-allowlist / WAF-skip recipe above. Say the word and I'll do them in a worktree off `origin/master`` like the doc flip.

164. user (2026-06-21T17:47:15.111Z)

Yes please! These two should be your last contributions for today! Appreciate the due diligence!

165. assistant (2026-06-21T17:47:58.453Z)

On it ? bundling both into **one** PR (they're a single "crawl hygiene for khelsutra.guru" work item; the repo rule is one-PR-per-item, no stacking). Let me first refresh khelsutra master and find the right home for the runbook:

166. user (2026-06-21T17:48:03.532Z)

=== origin/master tip ===

bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling tracked (#301) (#69)

=== site/ layout ===

```
README.md
access.db
make_og.py
node_modules
package-lock.json
package.json
public
schema.sql
scripts
server.js
```

```
src
wrangler.toml

=== site README / docs present? ===
site/README.md
--- repo docs/ (where deploy/site docs live) ---
README.md
--- deploy/ ---
DEPLOY_ALPHA.md
cloudflared.config.example.yml
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

167. assistant (2026-06-21T17:48:07.762Z)

`site/README.md` is the right home for the runbook. Let me read it to slot the section in cleanly:

168. user (2026-06-21T17:48:07.995Z)

```
1      # site/ ? public marketing site + Request-Access API (Cloudflare Worker)
2
3      The public site at **[khealsutra.guru](https://khealsutra.guru)** ? a static landing page
+
4      `/capture` checklist ? plus a small Worker that backs the Request-Access form with
**Cloudflare D1**.
5
6      ## Layout
7      - `public/` ? static assets served as-is (`index.html`, `capture.html`, `og.png`).
8      - `src/index.js` ? the Worker: serves `public/` and handles `POST /api/request-access` ?
D1.
9      - `src/index.test.js` ? unit tests for the form logic.
10     - `wrangler.toml` ? Worker + static-assets + D1 binding config.
11     - `schema.sql` ? the `access_requests` D1 table.
12     - `make_og.py` ? regenerates `public/og.png` (Pillow + Windows fonts): `python
make_og.py`.
13
14     ## Develop / test
15     ```
16     npm test      # node --test src/ (no install needed)
17     npm run dev   # npx wrangler dev ? local Worker + D1
18     ```
19
20     ## Deploy (owner ? our Cloudflare account)
21     One-time:
22     ```
23     npx wrangler d1 create khealsutra-access          # paste database_id
into wrangler.toml
24     npx wrangler d1 execute khealsutra-access --remote --file=schema.sql
25     ```
26     Then either:
27     - **Manual:** `npx wrangler deploy`, or
28     - **Git (recommended):** connect this repo in Cloudflare ? Workers Builds (root dir
`site/`), bind the
29     D1 database ? every merge to `master` deploys; PRs get preview URLs.
30
31     ## Notes
32     - The form posts to `/api/request-access`; if the Worker is unreachable it falls back to
a prefilled
33     `mailto:hello@`, so it degrades gracefully.
34     - `hello@khealsutra.guru` relies on Cloudflare Email Routing.
35     - The Web Analytics beacon in `public/index.html` has a placeholder token
(`REPLACE_WITH_CF_TOKEN`)
36     ? paste the real token or enable Cloudflare automatic setup.
37
```

169. assistant (2026-06-21T17:48:15.881Z)

Now I'll set up a worktree off `origin/master` and make both changes. Creating the worktree:

170. user (2026-06-21T17:48:18.226Z)

```
branch 'chore/site-crawl-hygiene' set up to track 'origin/master'.
HEAD is now at bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine
#299), scheduling tracked (#301) (#69)
=== worktree HEAD (origin/master bea8a2a) ===
chore/site-crawl-hygiene
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301) (#69)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

171. assistant (2026-06-21T17:48:31.172Z)

Now writing `robots.txt` and reading the worktree README (to register it for editing):

172. assistant (2026-06-21T17:48:34.534Z)

robots.txt ? khelsutra.guru (Cloudflare Worker: static site + Request-Access form API)

**Crawling the marketing pages is welcome. The only non-asset path is the form API
(`/api/request-access`, POST-only ? returns 405 to GET), which is not a crawl target.**

**NOTE: edge bot protection (Cloudflare Bot Fight Mode / WAF / rate limiting) runs in
FRONT of this file and can still 403 automated clients regardless of these rules ?
robots.txt is advisory, not enforcement. To crawl your OWN site through that layer,
allowlist the crawler in Cloudflare (see site/README.md ? "Crawling / bot protection")**

```
User-agent: *
Allow: /
Disallow: /api/
```

**Advisory pacing for crawlers that honor it (Googlebot ignores Crawl-delay and paces
itself). Keep automated traffic gentle ? this is a single small Worker.**

```
Crawl-delay: 1
```

173. user (2026-06-21T17:48:34.562Z)

File created successfully at: C:\Users\avidu\Projects\khelsutra-crawl-wt\site\public\robots.txt
(file state is current in your context ? no need to Read it back)

174. user (2026-06-21T17:48:39.707Z)

```
6      ## Layout
7      - `public/` ? static assets served as-is (`index.html`, `capture.html`, `og.png`).
8      - `src/index.js` ? the Worker: serves `public/` and handles `POST /api/request-access` ?
D1.
9      - `src/index.test.js` ? unit tests for the form logic.
10     - `wrangler.toml` ? Worker + static-assets + D1 binding config.
11     - `schema.sql` ? the `access_requests` D1 table.
12     - `make_og.py` ? regenerates `public/og.png` (Pillow + Windows fonts): `python
make_og.py`.
13
```

175. assistant (2026-06-21T17:48:57.733Z)

Now updating the README ? adding `robots.txt` to the Layout list and appending the crawling runbook:

176. user (2026-06-21T17:48:59.111Z)

The file C:\Users\avidu\Projects\khelsutra-crawl-wt\site\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

177. user (2026-06-21T17:49:13.161Z)

The file C:\Users\avidu\Projects\khelsutra-crawl-wt\site\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

178. assistant (2026-06-21T17:49:19.394Z)

Both changes in. Let me review the diff, then commit (explicit paths) and re-verify HEAD:

179. user (2026-06-21T17:49:21.091Z)

```
=== status ===
M site/README.md
?? site/public/robots.txt
```

```
=== new robots.txt ===
```

robots.txt ? khelsutra.guru (Cloudflare Worker: static site + Request-Access form API)

Crawling the marketing pages is welcome. The only non-asset path is the form API (`/api/request-access`, POST-only ? returns 405 to GET), which is not a crawl target.

NOTE: edge bot protection (Cloudflare Bot Fight Mode / WAF / rate limiting) runs in FRONT of this file and can still 403 automated clients regardless of these rules ? robots.txt is advisory, not enforcement. To crawl your OWN site through that layer, allowlist the crawler in Cloudflare (see site/README.md ? "Crawling / bot protection")

```
User-agent: *
Allow: /
Disallow: /api/
```

Advisory pacing for crawlers that honor it (Googlebot ignores Crawl-delay and paces itself). Keep automated traffic gentle ? this is a single small Worker.

```
Crawl-delay: 1
```

```
=== README diff ===
diff --git a/site/README.md b/site/README.md
index bd956c1..a793400 100644
--- a/site/README.md
+++ b/site/README.md
@@ -4,7 +4,7 @@ The public site at [khelsutra.guru](https://khelsutra.guru) ? a static lan
   `capture` checklist ? plus a small Worker that backs the Request-Access form with Cloudflare D1.
```

```
## Layout
-- `public/` ? static assets served as-is (`index.html`, `capture.html`, `og.png`).
+ `public/` ? static assets served as-is (`index.html`, `capture.html`, `og.png`,
`robots.txt`).
- `src/index.js` ? the Worker: serves `public/` and handles `POST /api/request-access` ? D1.
```

```

- `src/index.test.js` ? unit tests for the form logic.
- `wrangler.toml` ? Worker + static-assets + D1 binding config.
@@ -34,3 +34,40 @@ Then either:
- `hello@khelsutra.guru` relies on Cloudflare Email Routing.
- The Web Analytics beacon in `public/index.html` has a placeholder token
(`REPLACE_WITH_CF_TOKEN`)
  ? paste the real token or enable Cloudflare automatic setup.
+
+### Crawling / bot protection
+
+`khelsutra.guru` sits behind Cloudflare's edge security (Bot Fight Mode / WAF / rate
+limiting), which runs in front of the Worker. The app itself never returns `403` ?
+`src/index.js` only emits `405`/`400` on the form path and `200`/`404` for assets ? so
+a `403` while crawling is Cloudflare's edge, not this site. `public/robots.txt` is
+advisory (and won't change a `403`); the edge mitigation is the thing to adjust.
+
+#### Diagnose which Cloudflare feature is blocking
+From the crawler's host (prints status + Cloudflare headers):
+```
+curl -sS -D - -o /dev/null -A "<crawler UA>" "https://khelsutra.guru/<failing path>"
+```
+- `cf-mitigated: challenge` (or a "Checking your browser" body) ? Bot Fight Mode
+- body "Sorry, you have been blocked" + a Ray ID ? WAF rule
+- HTTP `429`, or `403` + a `Retry-After` header ? Rate limiting
+
+Authoritative: Cloudflare dashboard ? khelsutra.guru ? Security ? Events, filter
+Action = Block/Challenge ? each row names the blocking service + the Ray ID.
+
+#### Let your OWN crawler through (allowlist; don't weaken protection globally)
+1. IP Access Rule = Allow (Security ? WAF ? Tools ? IP Access Rules) for the crawler's
+ egress IP (`curl -s https://api.ipify.org` from the crawler). Bypasses Bot Fight Mode +
+ security level; works on the free plan ? the simplest reliable fix.
+2. WAF Skip rule by secret header (Pro+): custom rule
+  `any(http.request.headers["x-crawl-key"][*] eq "<secret>")` ? Action Skip ? Super Bot
+  Fight Mode + managed rules + rate limiting; have the crawler send `X-Crawl-Key: <secret>`.
+  Robust regardless of IP/UA. (Plain Bot Fight Mode on the free plan can't be selectively
+  skipped ? use #1 or toggle it off.)
+3. Toggle Bot Fight Mode off (Security ? Bots) for a short crawl window.
+4. Rate-limited? Lower concurrency + add a delay (~1 req/s); a burst of identical-looking
+  requests trips heuristics even with a good UA.
+
+Crawler-side (no Cloudflare change): send a real browser `User-Agent` + `Accept` /
+`Accept-Language`. Light crawling may pass, but Bot Fight Mode fingerprints beyond the UA,
+so #1/#2 are the durable fixes.
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

180. assistant (2026-06-21T17:49:30.633Z)

Clean diff. Committing (explicit paths) and verifying lineage:

181. user (2026-06-21T17:49:36.675Z)

```

=== re-verify HEAD (mine, off origin/master) ===
chore/site-crawl-hygiene
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301) (#69)
[chore/site-crawl-hygiene 37a3f0d] chore(site): add robots.txt + a "crawling / bot protection"
runbook for khelsutra.guru
 2 files changed, 55 insertions(+), 1 deletion(-)
 create mode 100644 site/public/robots.txt
=== only my commit ahead of origin/master ===
37a3f0d chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

182. assistant (2026-06-21T17:49:40.279Z)

Pushing and opening the PR:

183. user (2026-06-21T17:49:53.381Z)

```
branch 'chore/site-crawl-hygiene' set up to track 'origin/chore/site-crawl-hygiene'.
To https://github.com/avidullu/khelsutra.git
* [new branch]      chore/site-crawl-hygiene -> chore/site-crawl-hygiene
=== create PR ===
https://github.com/avidullu/khelsutra/pull/71
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

184. assistant (2026-06-21T17:49:56.648Z)

PR ****[#71](https://github.com/avidullu/khelsutra/pull/71)**** opened. Watching CI to green, then I'll merge:

185. user (2026-06-21T17:50:45.447Z)

```
site    pass    13s
https://github.com/avidullu/khelsutra/actions/runs/27912565182/job/82592118195
web      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27912565182/job/82592118183
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/80b12a66-7b41-4099-a58f-71b97214b17f
site    pass    13s
https://github.com/avidullu/khelsutra/actions/runs/27912565182/job/82592118195
web      pass    35s
https://github.com/avidullu/khelsutra/actions/runs/27912565182/job/82592118183
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/80b12a66-7b41-4099-a58f-71b97214b17f
site    pass    13s
https://github.com/avidullu/khelsutra/actions/runs/27912565182/job/82592118195
web      pass    35s
https://github.com/avidullu/khelsutra/actions/runs/27912565182/job/82592118183
=== watch exit=0 ===
```

186. assistant (2026-06-21T17:50:49.387Z)

All green. Squash-merging and cleaning up:

187. user (2026-06-21T17:50:59.692Z)

```
=== PR state ===
{"mergedAt":"2026-06-21T17:50:54Z","state":"MERGED"}
=== cleanup: worktree + local branch + remote branch ===
Deleted branch chore/site-crawl-hygiene (was 37a3f0d).
- [deleted]      chore/site-crawl-hygiene
=== confirm master has it ===
From https://github.com/avidullu/khelsutra
   bea8a2a..f7e2bd5  master    -> origin/master
f7e2bd5 chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru (#71)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

188. assistant (2026-06-21T17:51:03.461Z)

Merged into khelsutra master (`f7e2bd5``). Let me confirm the final clean state across both repos:

189. user (2026-06-21T17:51:06.376Z)


```
=== khelsutra worktrees (crawl-wt gone) ===
C:/Users/avidu/Projects/khelsutra          46e0244 [master]
C:/Users/avidu/Projects/khelsutra-b3-wt    5ac1c63 [feat/b3-site-i18n]
=== khelsutra main checkout (left as found) ===
master
?? .claude/
?? site/access.db

=== engine: worktrees (no stray t11/crawl) + branch ===
(no stray worktrees)
serving/exposure-safety-preflight
```

190. assistant (2026-06-21T17:51:28.735Z)

Both delivered and merged ? and everything's tidied up. ?

Shipped (one PR, since it's a single "crawl hygiene" work item)

```
**[khelsutra #71](https://github.com/avidullu/khelsutra/pull/71) ? MERGED** (`f7e2bd5`, all CI green):
- **`site/public/robots.txt`** ? `Allow: /`, `Disallow: /api/`, advisory `Crawl-delay`; with a note that robots.txt is *advisory* and the Cloudflare edge is the real knob.
- **`site/README.md` ? "Crawling / bot protection"*** ? a runbook to diagnose *which* Cloudflare feature blocks (the `curl -D -` header read + Security ? Events) and to allowlist your own crawler.
```

The actionable takeaway for your sibling crawl session

The 403 is **Cloudflare's edge bot protection**, not your site ? proven by elimination: your Worker never emits 403, and a plain `curl` pulled the homepage. So the fastest unblock right now:

1. From the crawler: `curl -s https://api.ipify.org` to get its egress IP.
2. Cloudflare ? **khelsutra.guru ? Security ? WAF ? Tools ? IP Access Rules** ? add that IP, action **Allow**.

That bypasses Bot Fight Mode + security level, works on the free plan, and keeps protection on for everyone else. If the IP isn't stable, the header-based WAF Skip rule (Pro+) in the runbook is the durable alternative. And if you peek at **Security ? Events** first, it'll name the exact service + Ray ID so you know whether you're fighting Bot Fight Mode, a WAF rule, or rate limiting.

Final state

```
- khelsutra `master` ? `f7e2bd5`; engine `master` ? `cb7394b` (T11). Both PRs (#69, #71) and the engine T11 PR (#299) merged; issue [#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301) tracks the sweep automation.
- All my worktrees removed, feature branches deleted (local + remote), and both main checkouts left exactly as I found them ? the UX session's `khelsutra-b3-wt` and the serving session's branch were never touched.
```

That's everything for today ? T11 implemented + merged, the doc flipped, the automation tracked, and your crawl 403 diagnosed with a fix in hand. Thanks for the great back-and-forth ? enjoy the rest of your day! ?